

LECTURE 1 ANATOMY

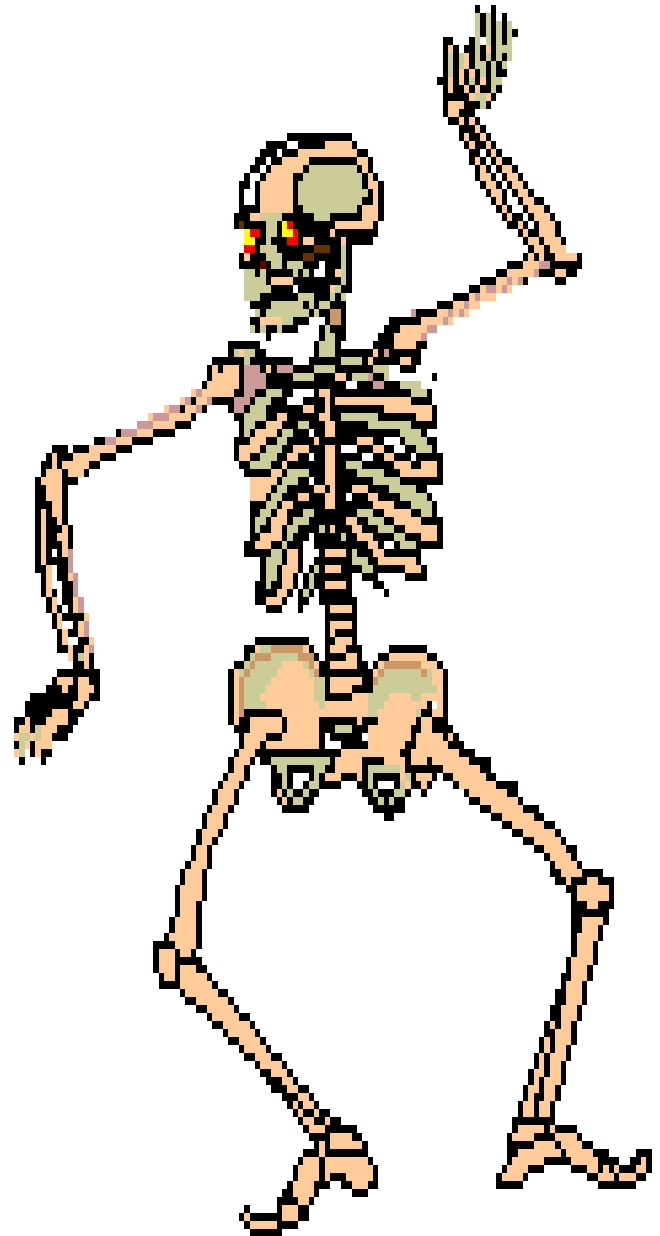
- Bones of the skeleton and skeletal system**
- Anatomical terms and movements**

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ILOs:

1. **List** bones forming the axial skeleton & bones forming the appendicular skeleton
2. **Classify** bones according to shape, structure & development giving examples.
3. **Describe** the structure of a long growing bone
4. **Explain** the different types of bone ossification .
5. **Illustrate** arterial supply of bones

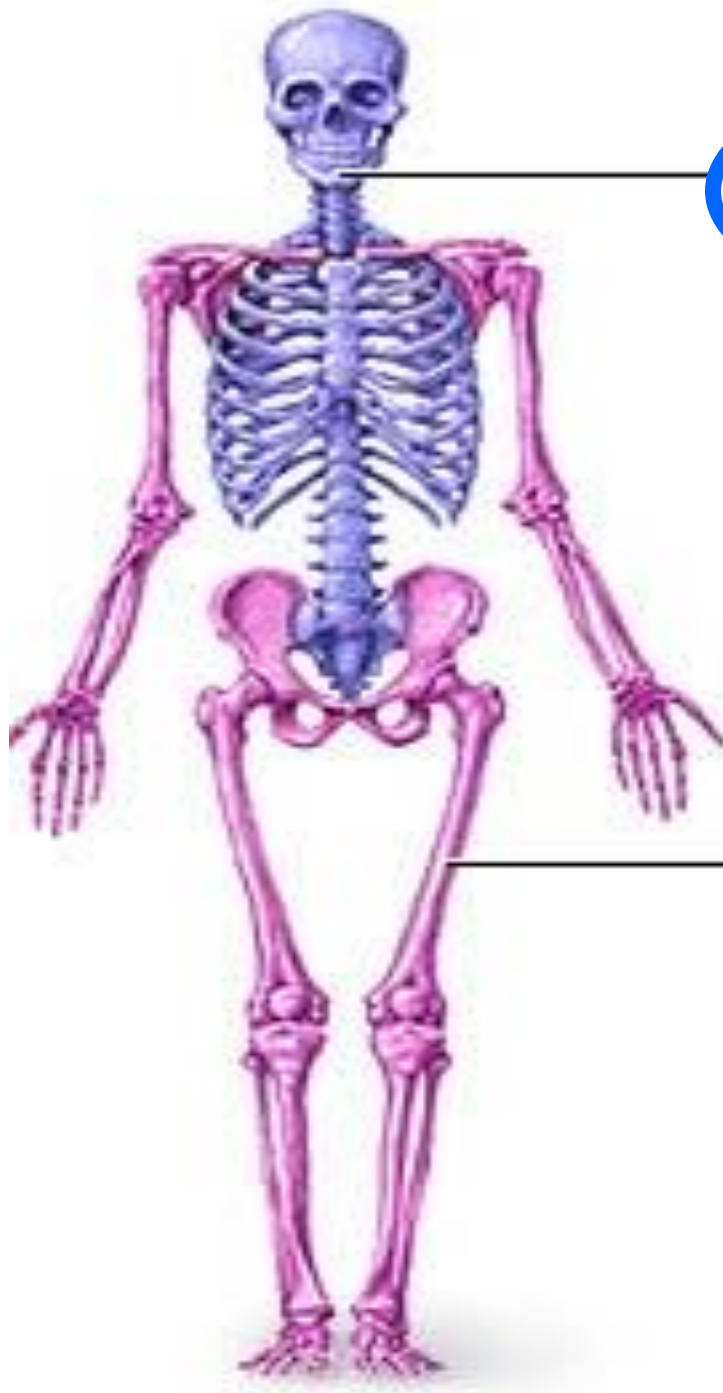
The Skeleton
is formed of
(206 bones)



Parts of the skeleton

- (1) **Axial skeleton**
→ bones in the
midline of the
body
- (2) **Appendicular
skeleton** → bones
of the limbs &
their **girdles**





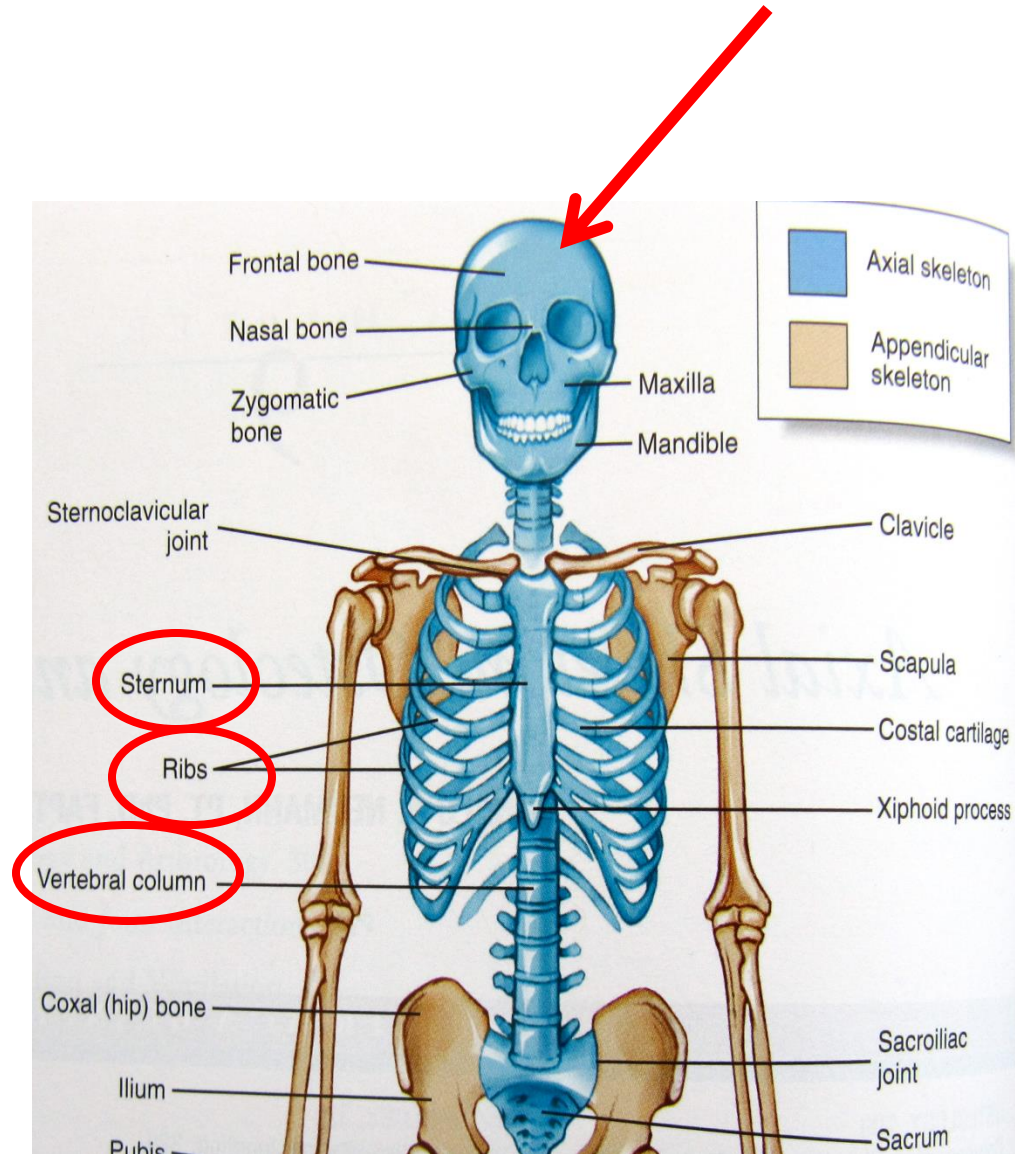
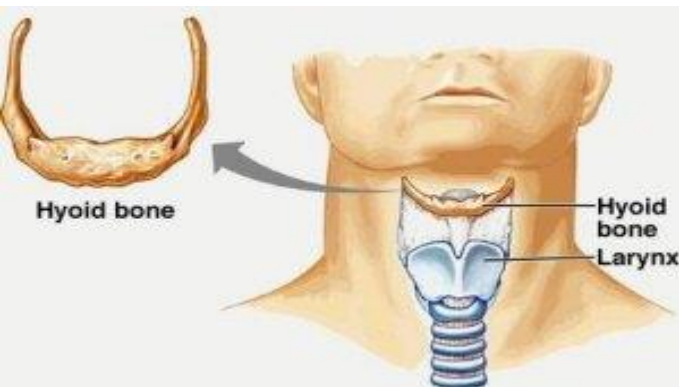
Axial skeleton

**Appendicular
skeleton**

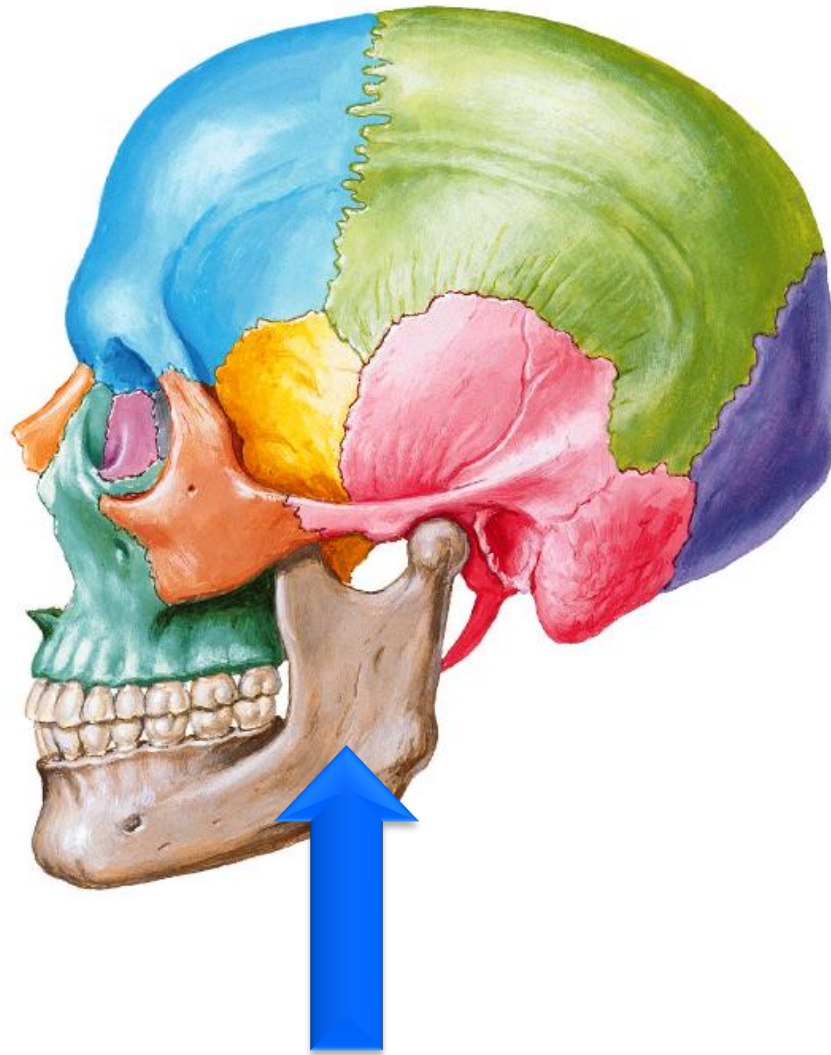
Axial skeleton

• Formed of:

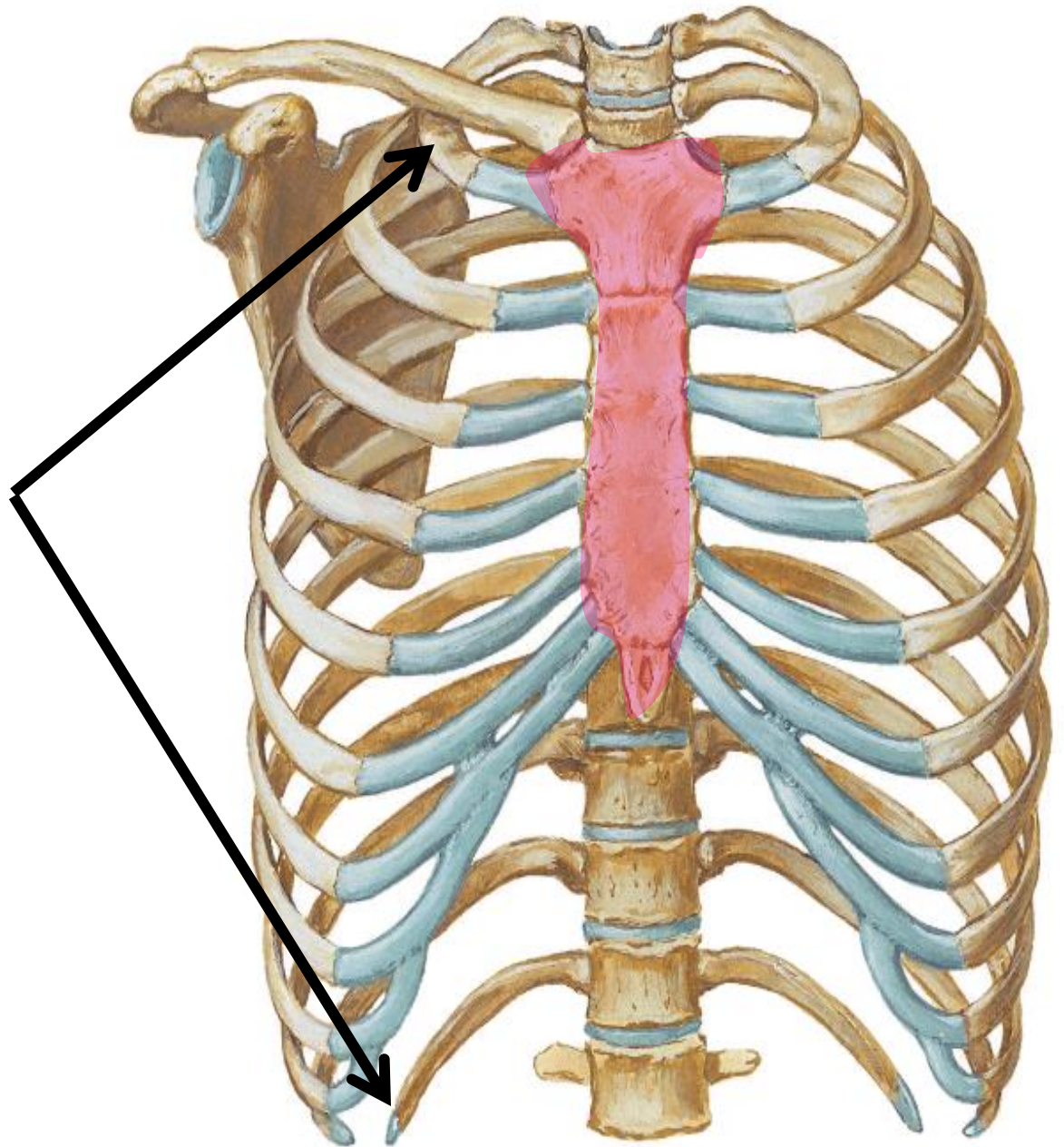
1. Skull
2. Ribs
3. Sternum
4. Vertebral column
5. Hyoid bone



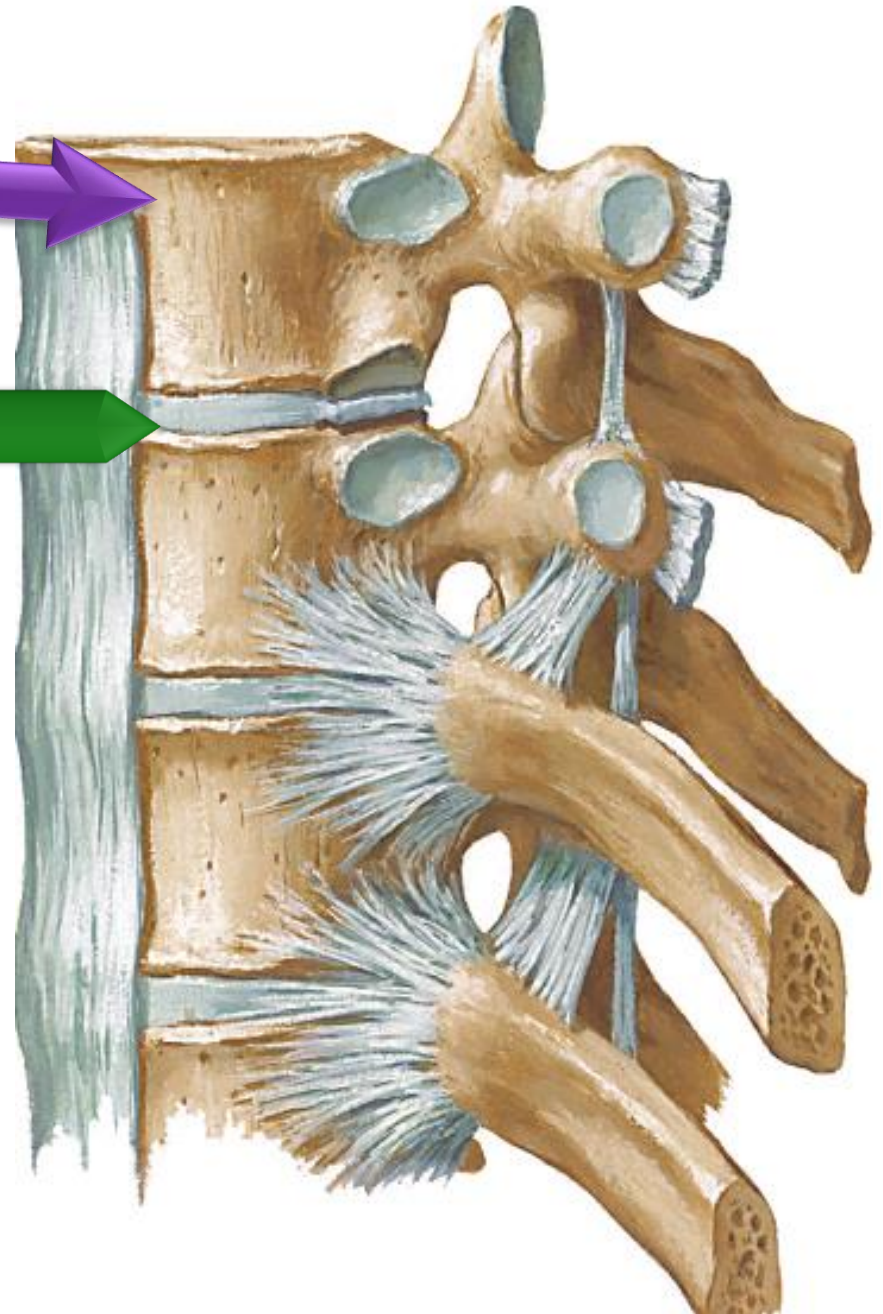
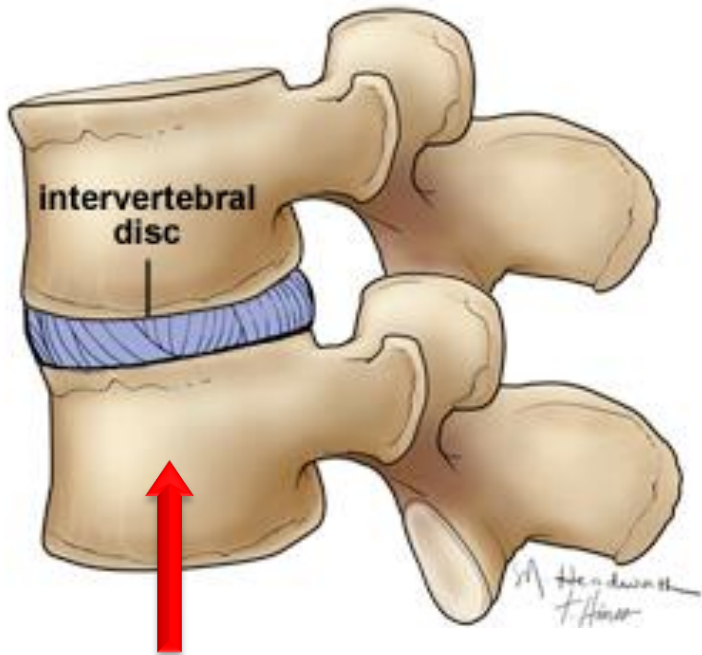
- **Skull**
-
- **Mandible**



- Ribs
- (12 pairs)
- Sternum

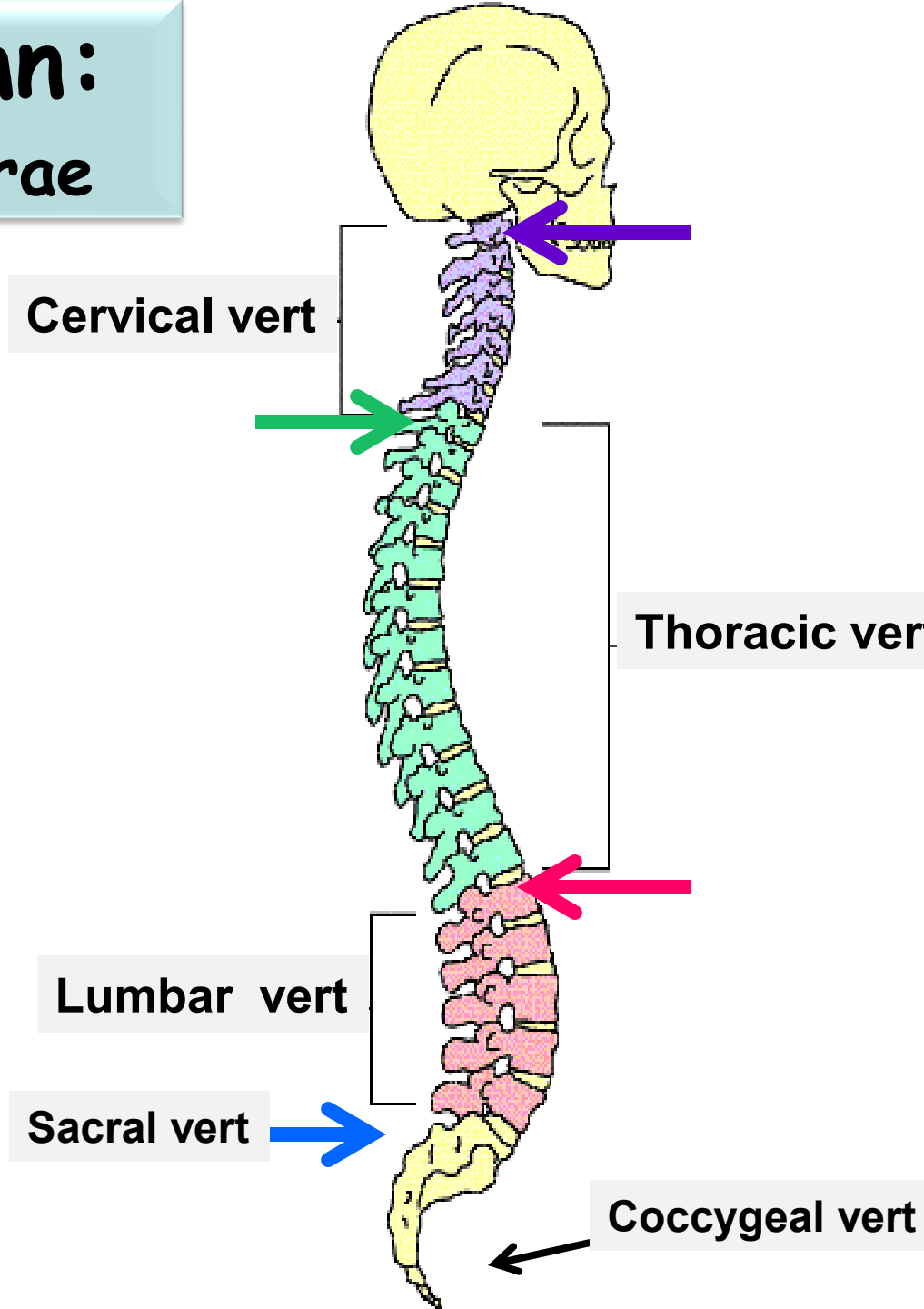


- Vertebra
- Intervertebral disc



Vertebral column: formed of **33** vertebrae

- **7 cervical vertebrae**
- **12 thoracic vertebrae**
- **5 lumbar vertebrae**
- **5 sacral vertebrae fused to form sacrum**
- **2-4 coccygeal vertebrae fused to form coccyx**



Appendicular Skeleton

Free upper limb
+ shoulder girdle

Free Lower limb
+ pelvic girdle

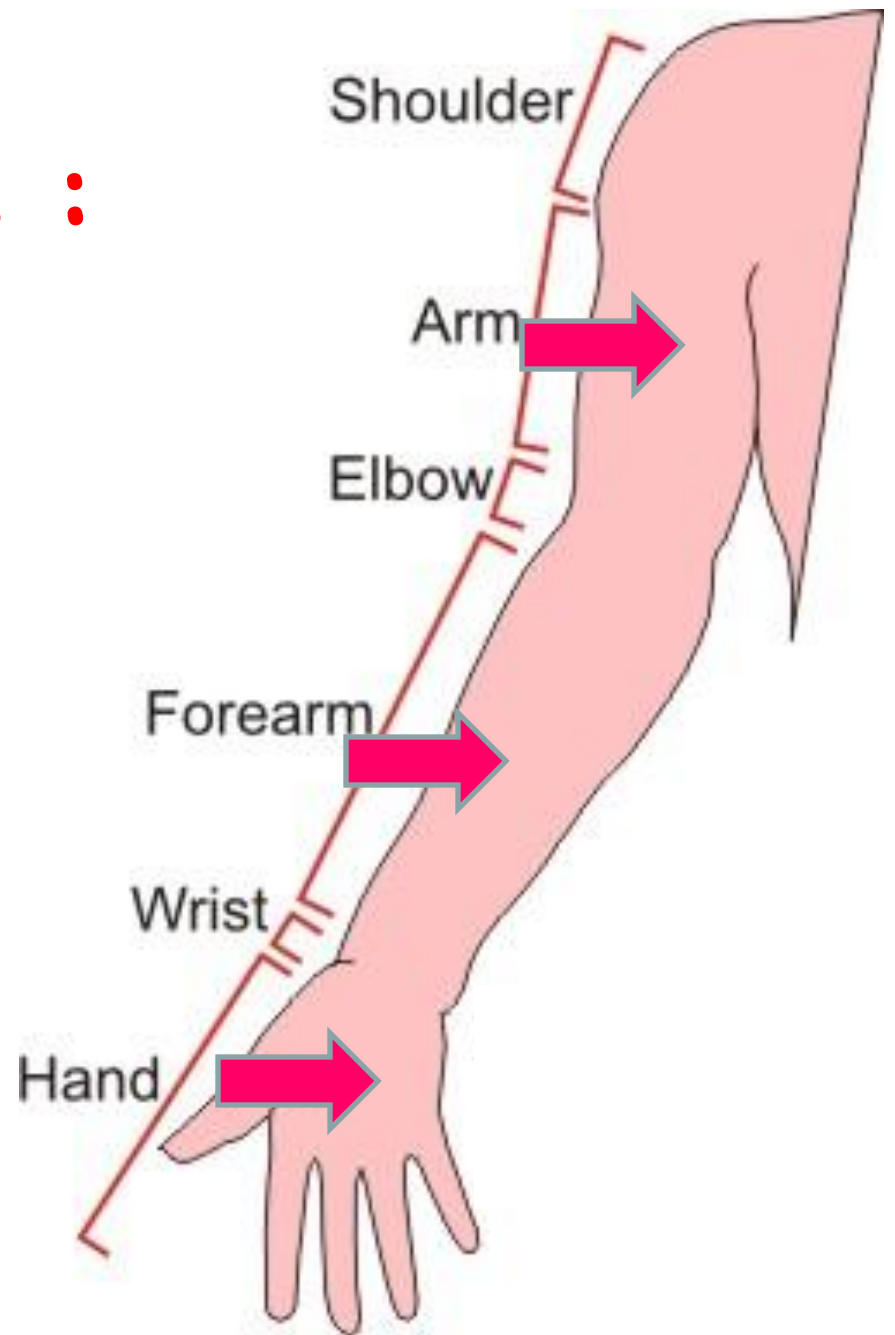


Upper limb parts :

Arm

Forearm

Hand

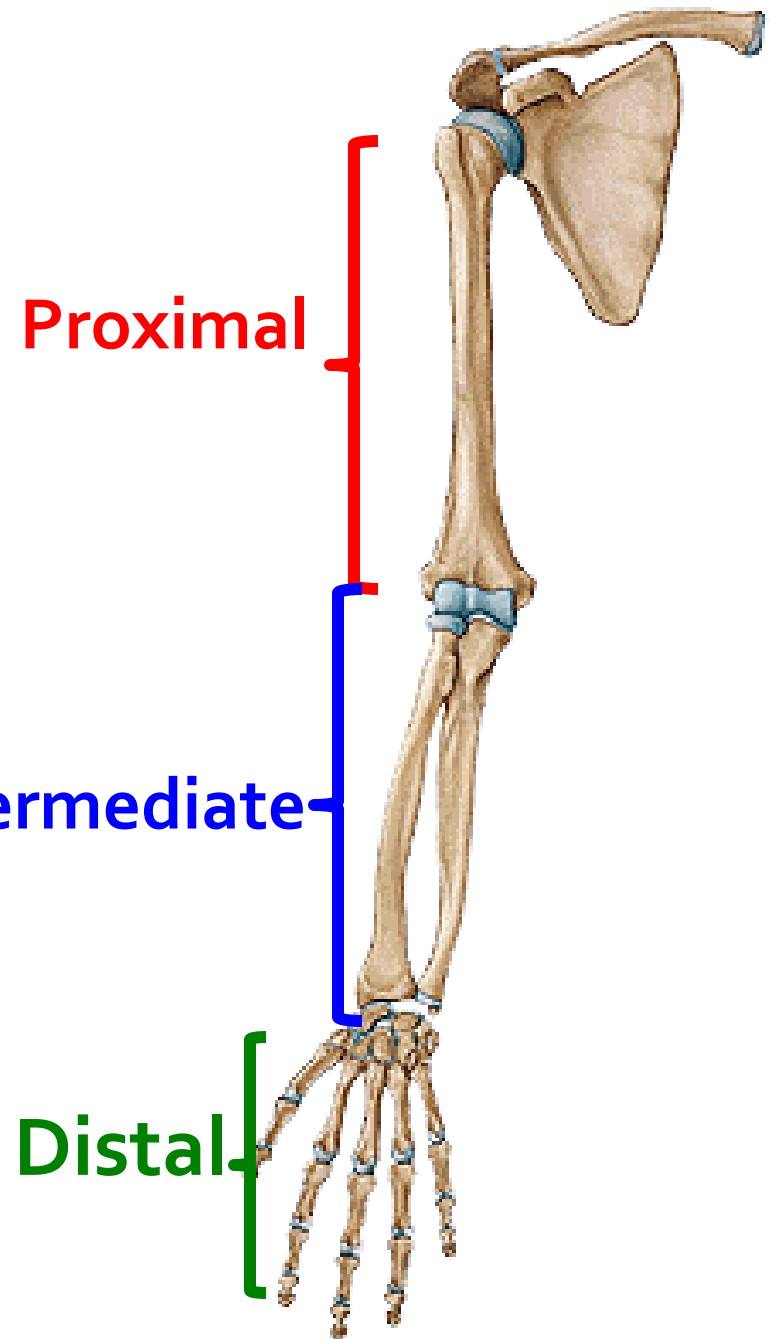


Appendicular skeleton:
Bones of free limbs are formed
of 3 segments:

Bone of arm → humerus

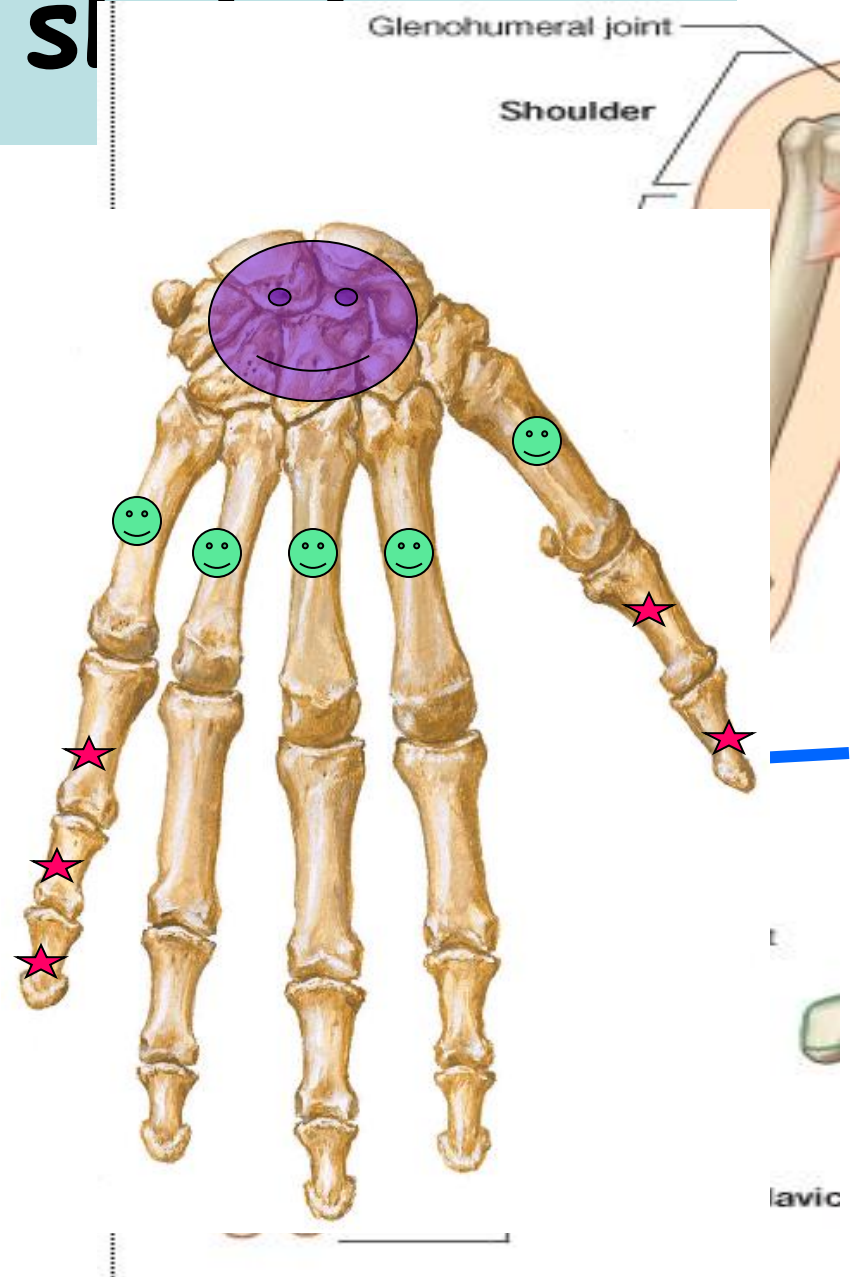
Bones of forearm → radius & ulna

Bones of hand → carpals
, metacarpals &
phalanges



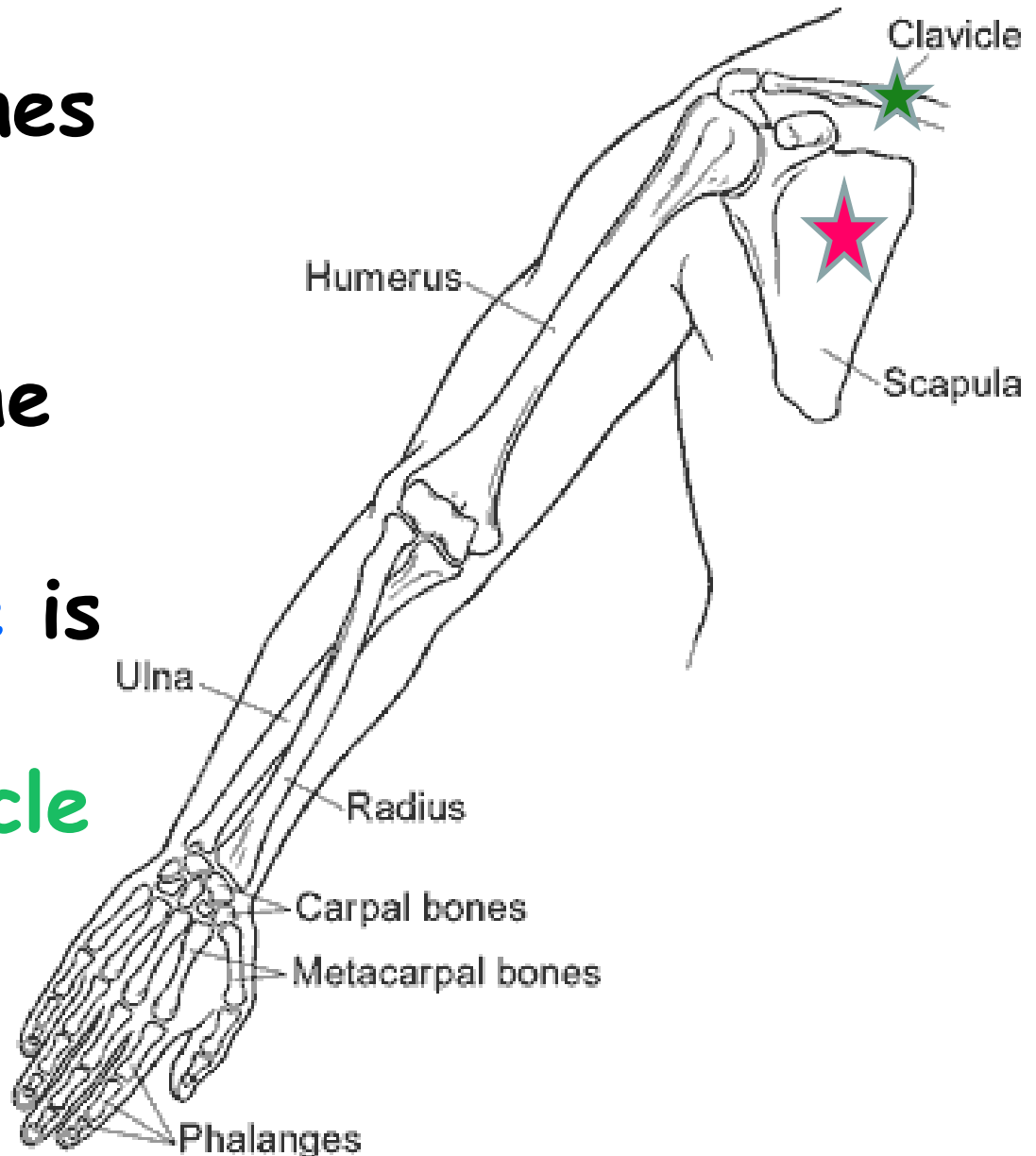
Appendicular skeleton

- (A) Bones of upper limb:
- Bone of arm → humerus
- Bones of forearm → radius & ulna
- Bones of hand → carpals (8), metacarpals (5) & phalanges (3) except thumb (2)

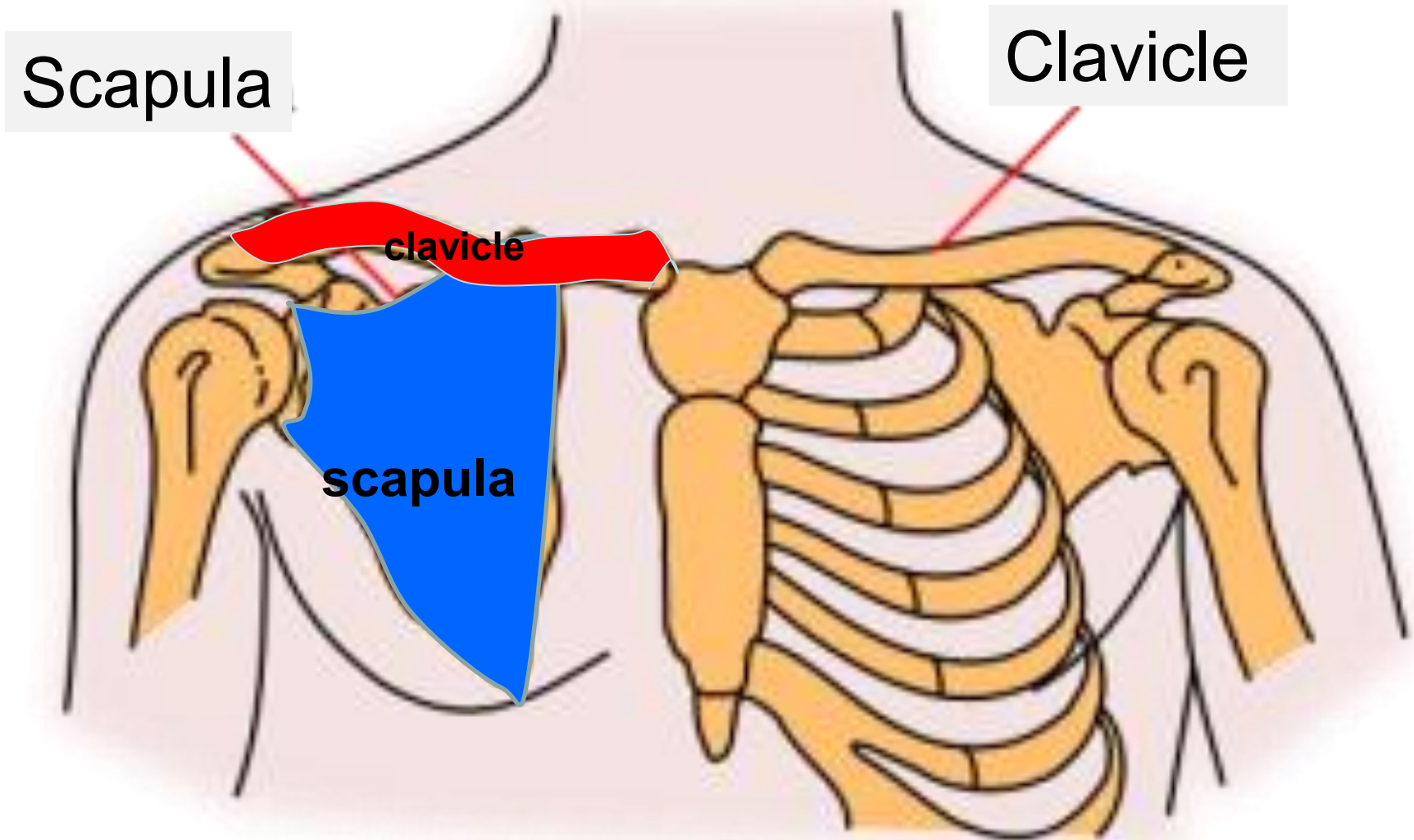


Bones of shoulder girdle

- Girdles are bones which connect bones of the free limb to the axial skeleton
- **Shoulder girdle** is formed of **scapula** & **clavicle**



Shoulder girdle

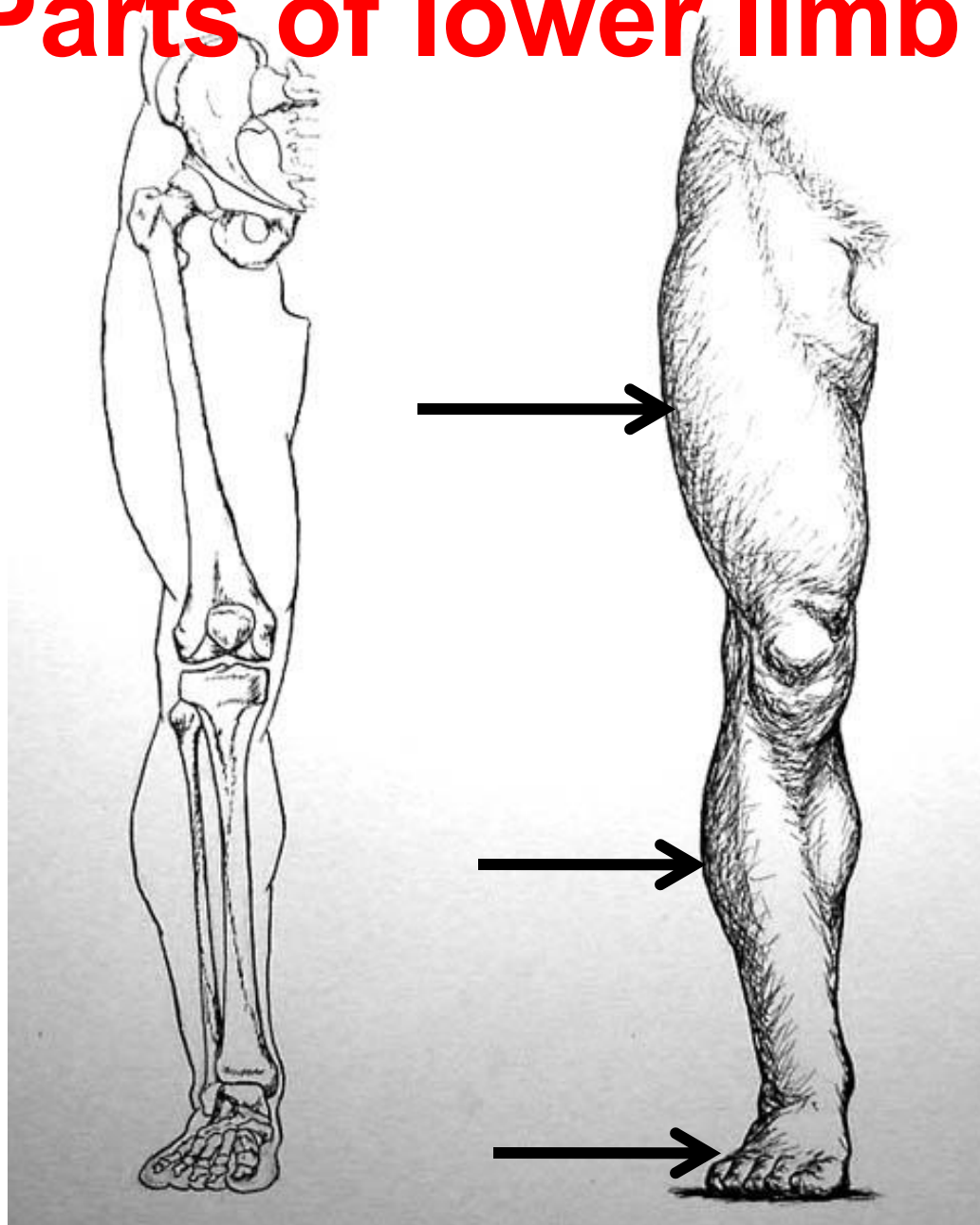


Parts of lower limb

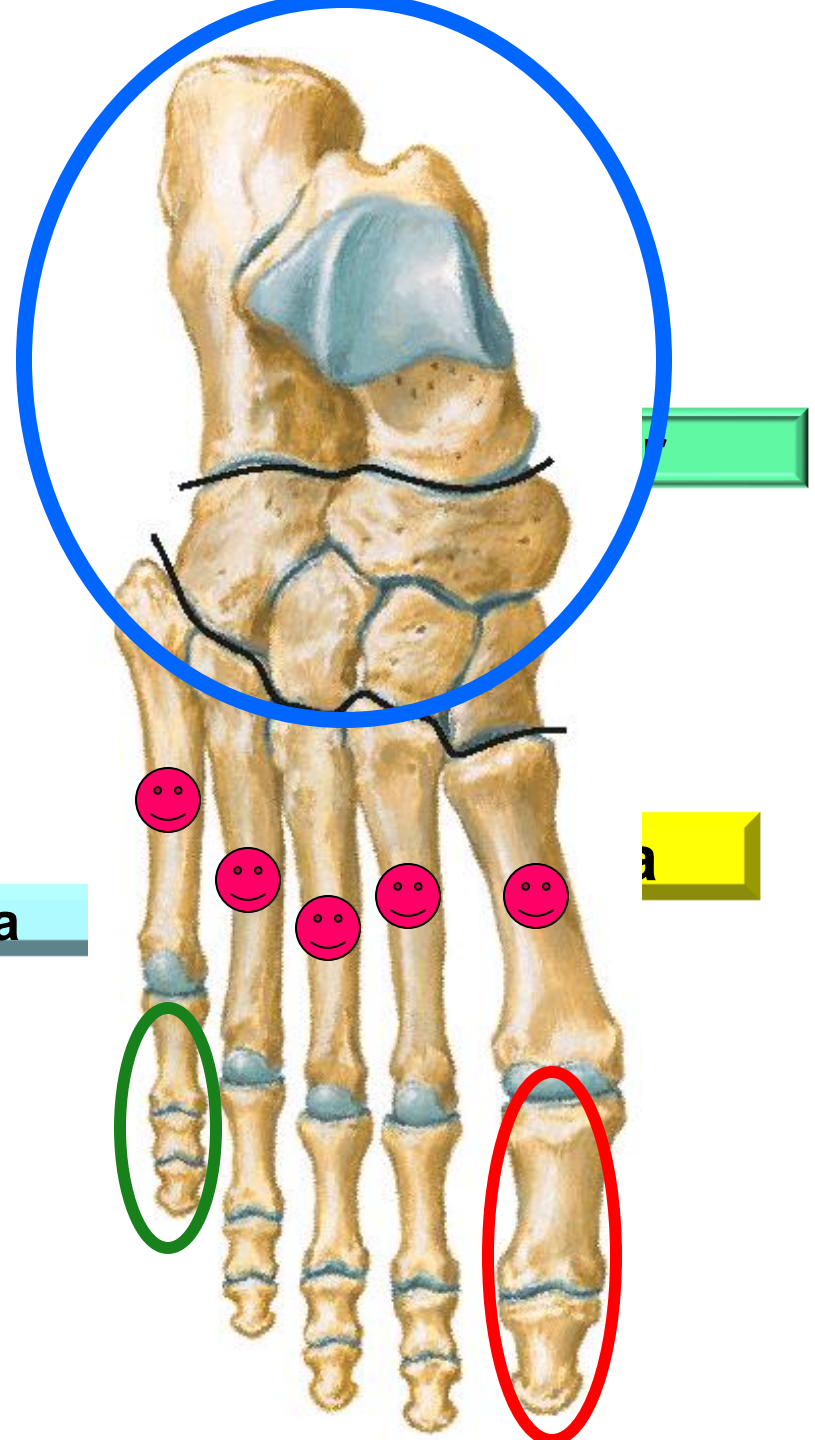
Thigh

Leg

Foot

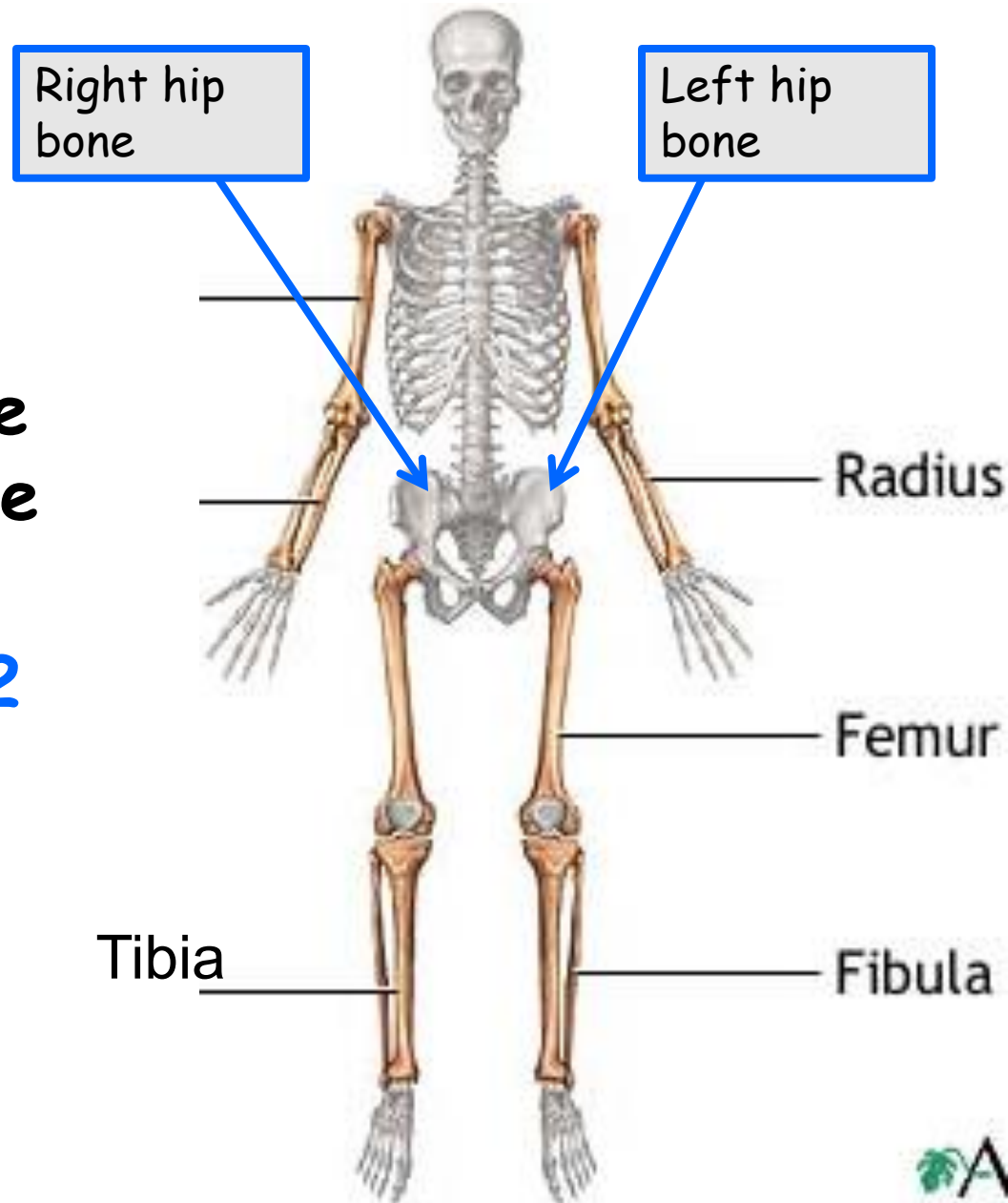


- (B) Bones of the lower limb:
- Bone of thigh → femur
- Bones of leg → tibia & fibula
- Bones of foot → tarsals (7), metatarsals (5) & phalanges (3) except big toe (2)

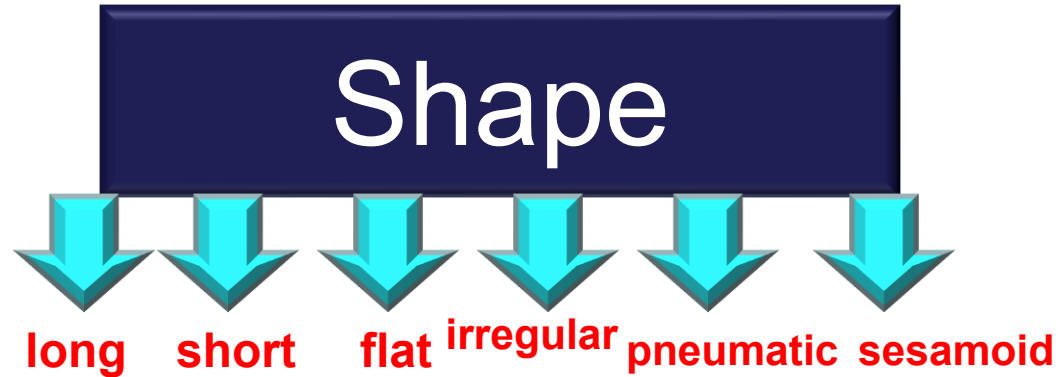
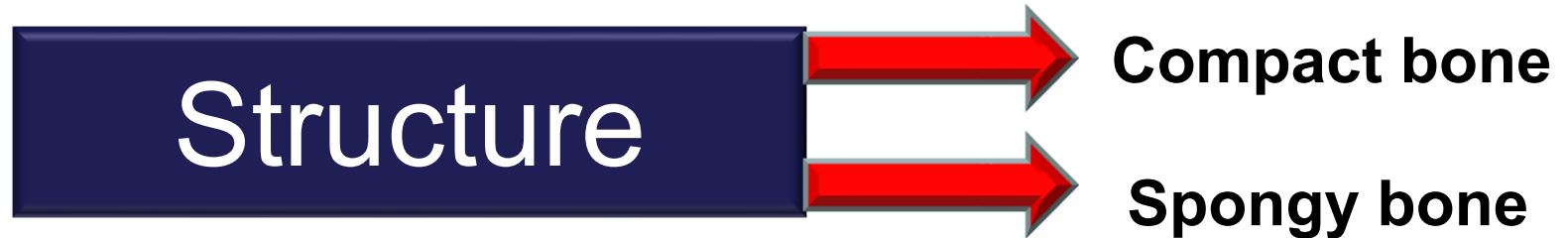


Pelvic girdle

- The **pelvic girdle** is the bone which connects bones of the free lower limb to the axial skeleton
- It is formed of the **2 hip bones**

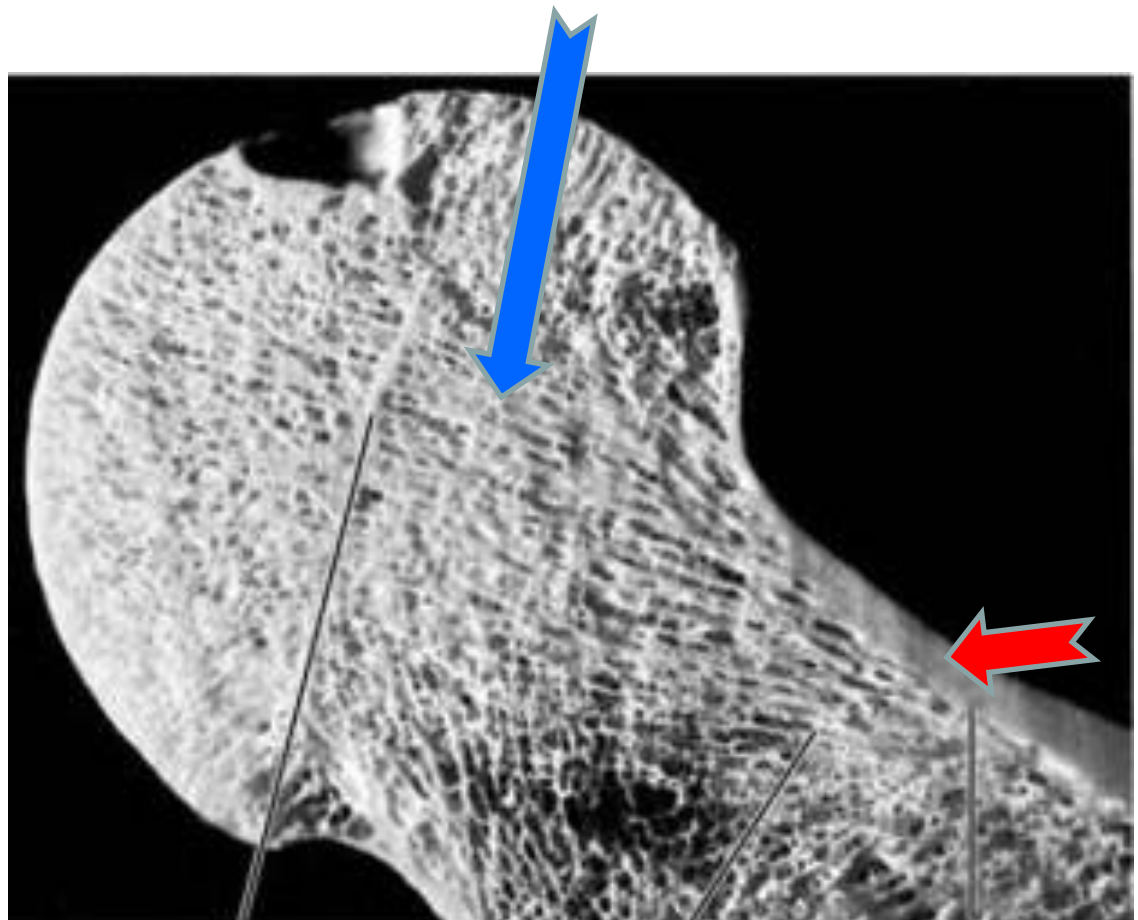
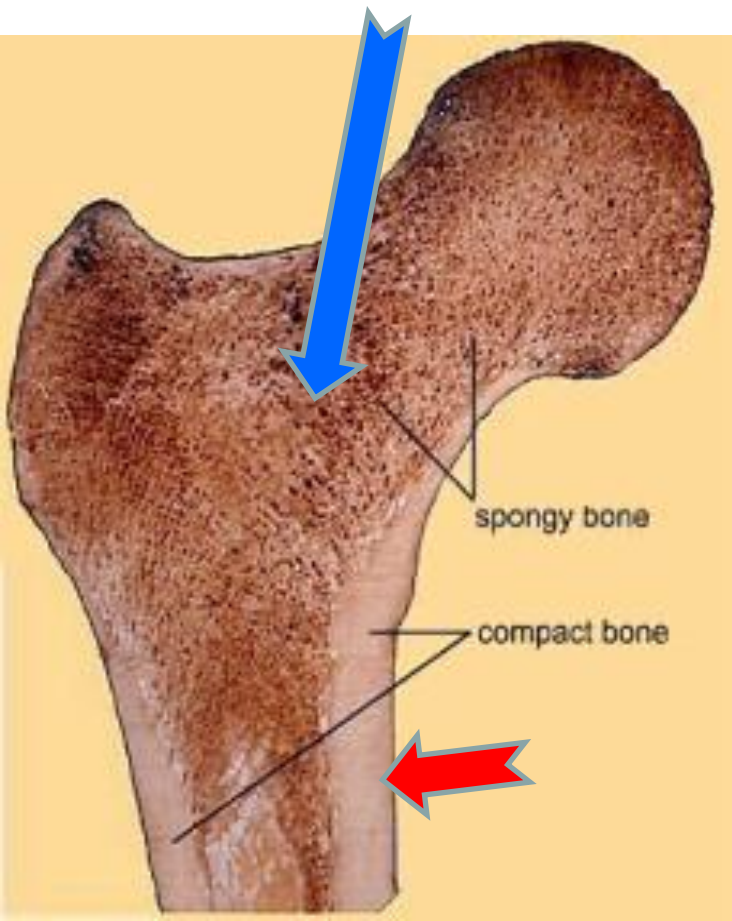


Types of Bones

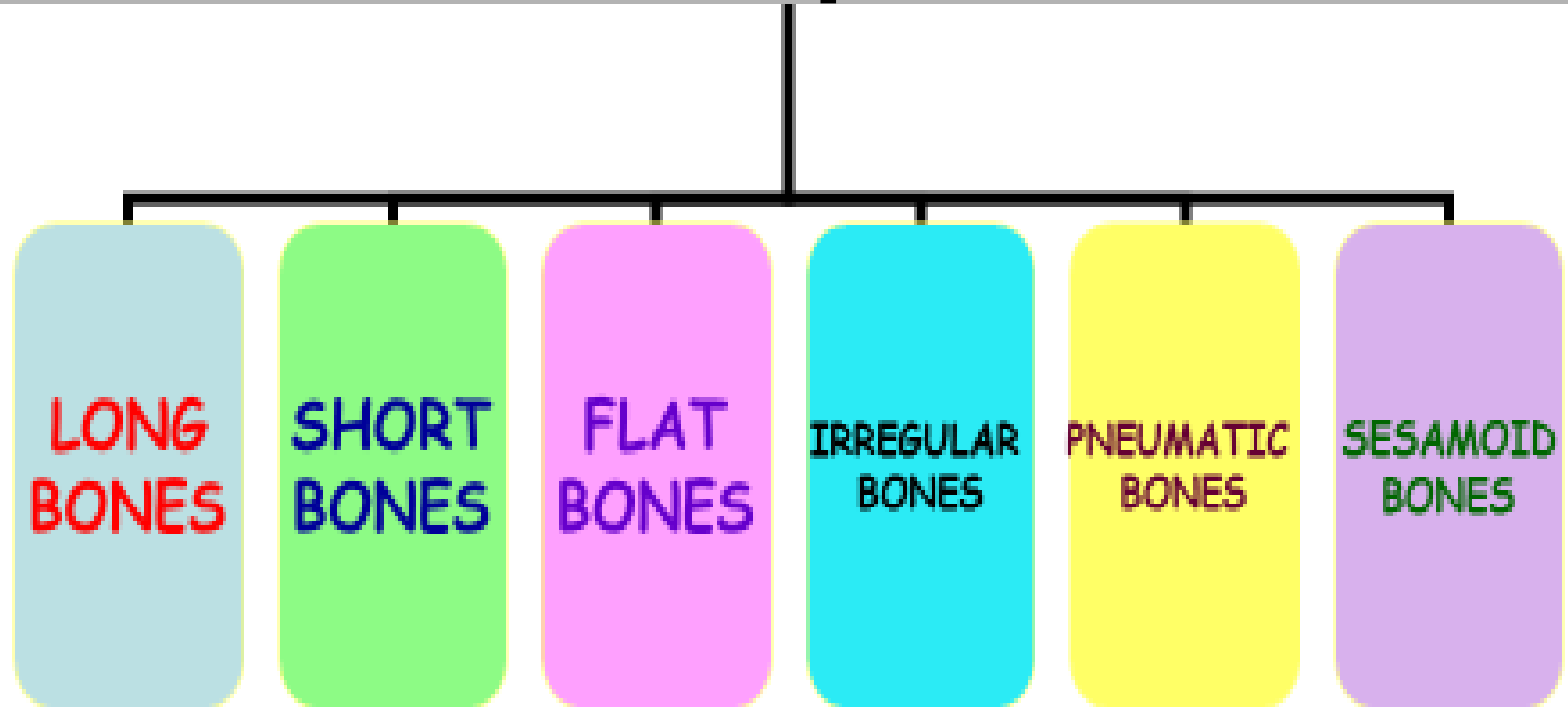


Types of bone according to Structure

Types of bone → **compact** (ivory like)
→ **cancellous** (spongy)

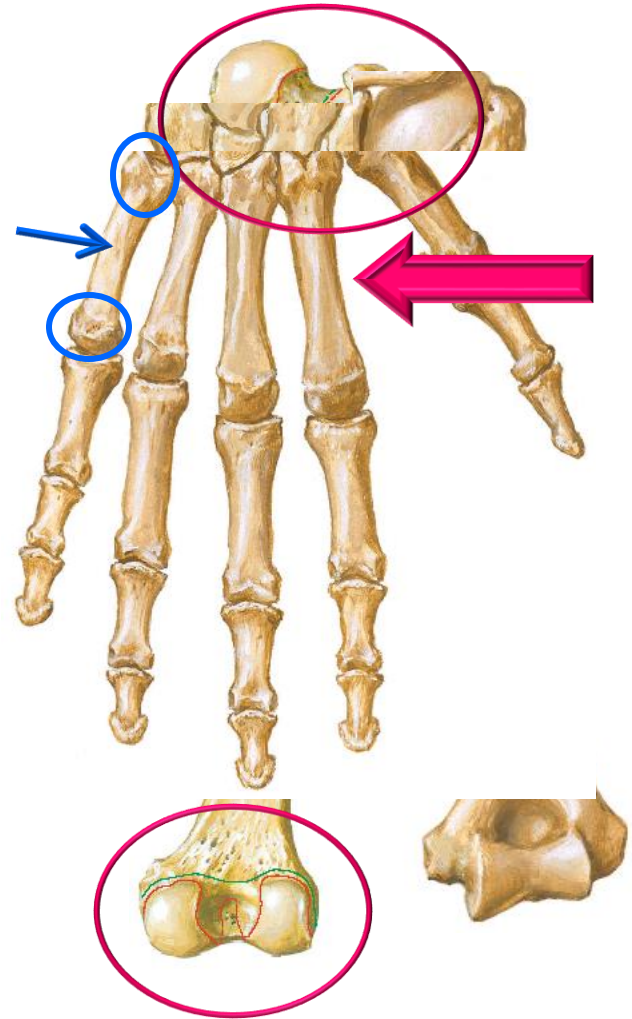


Types of bone according to Shape

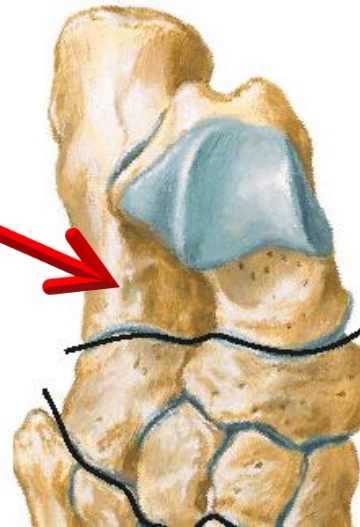
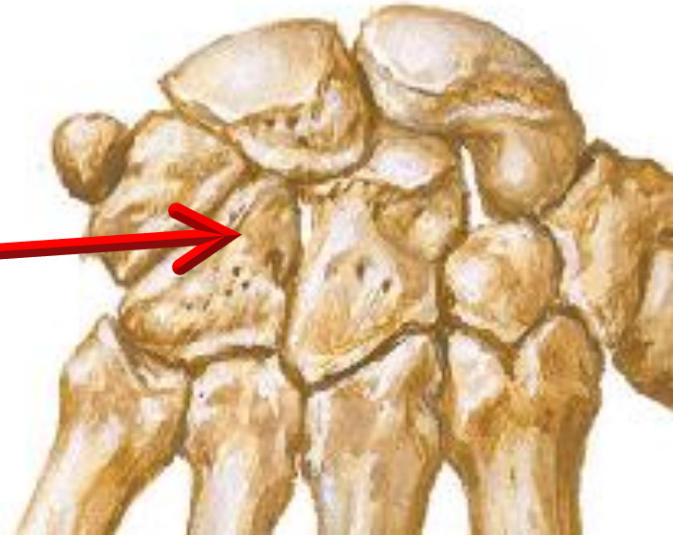


Types of bones according to shape

- 1) Long bones :
- (2 ends & a shaft)
→ femur, humerus
- The metacarpals, metatarsals, & phalanges are considered as small models of long bones as they have 2 ends & a shaft



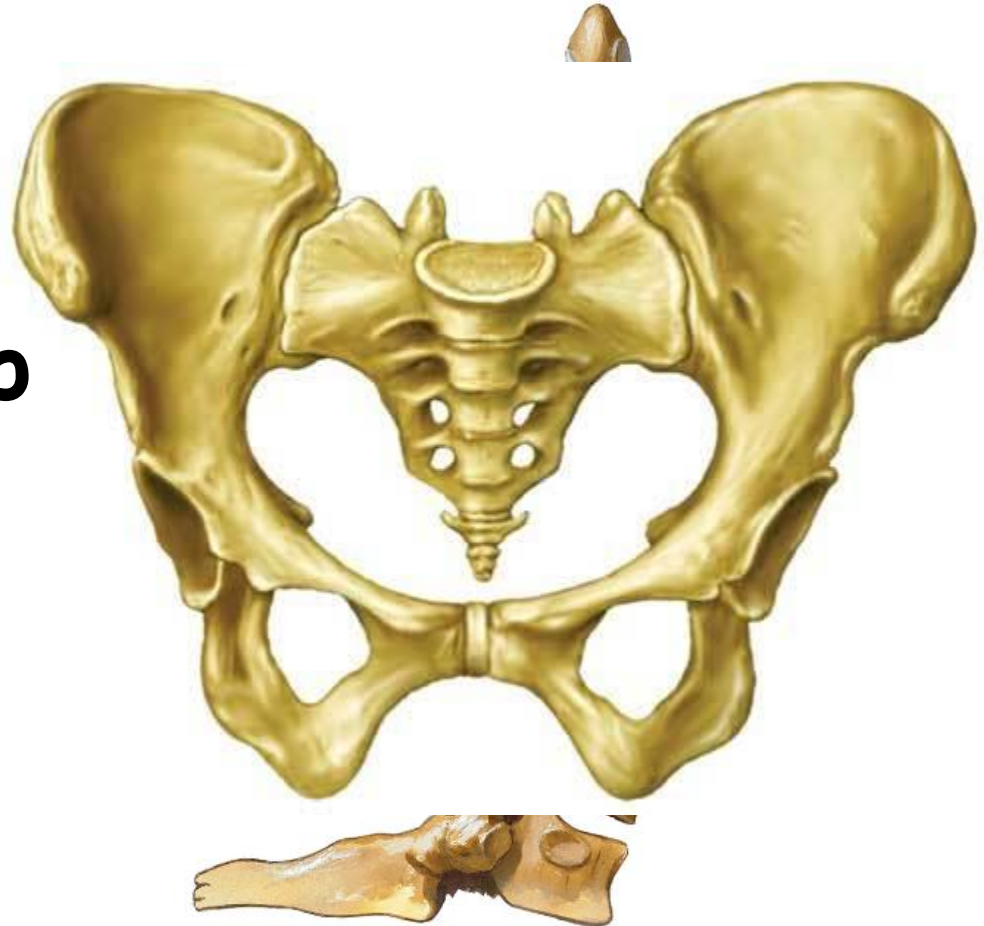
- 2) short bones
→ carpal & tarsal bones



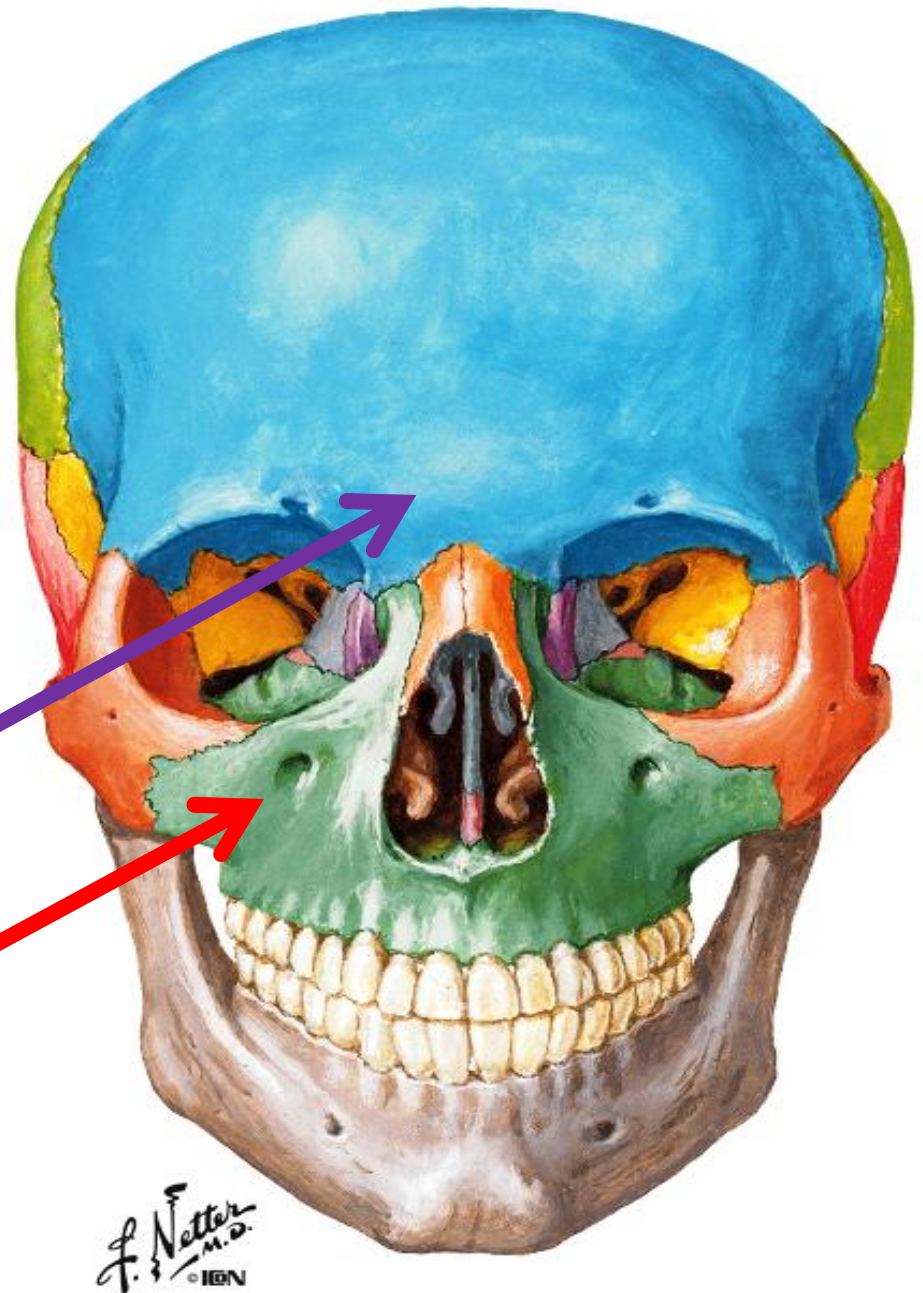
- 3) flat bones
→ scapula ,
sternum &
bones of skull
cap



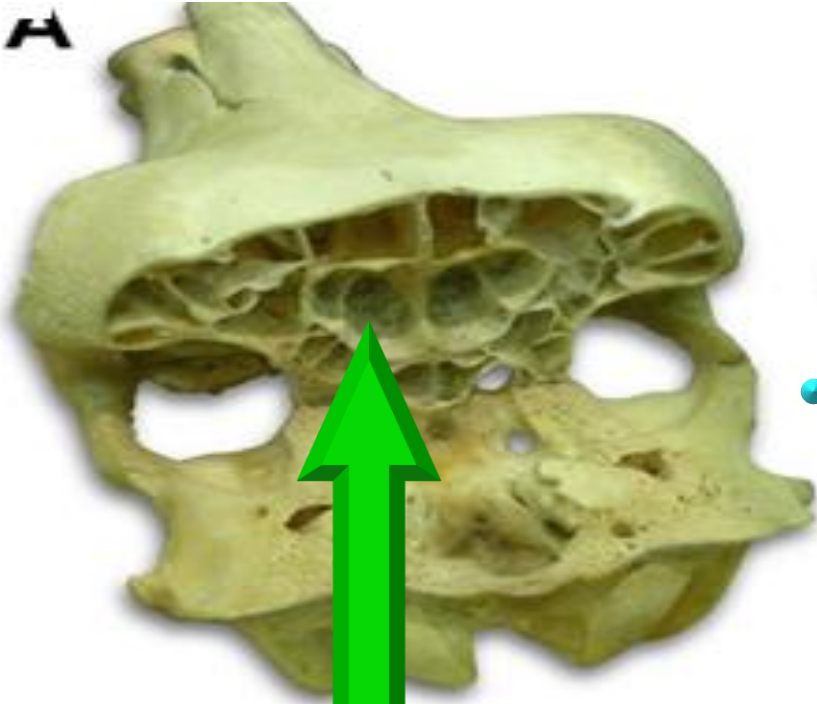
- 4) Irregular bones → vertebrae/ hip bone



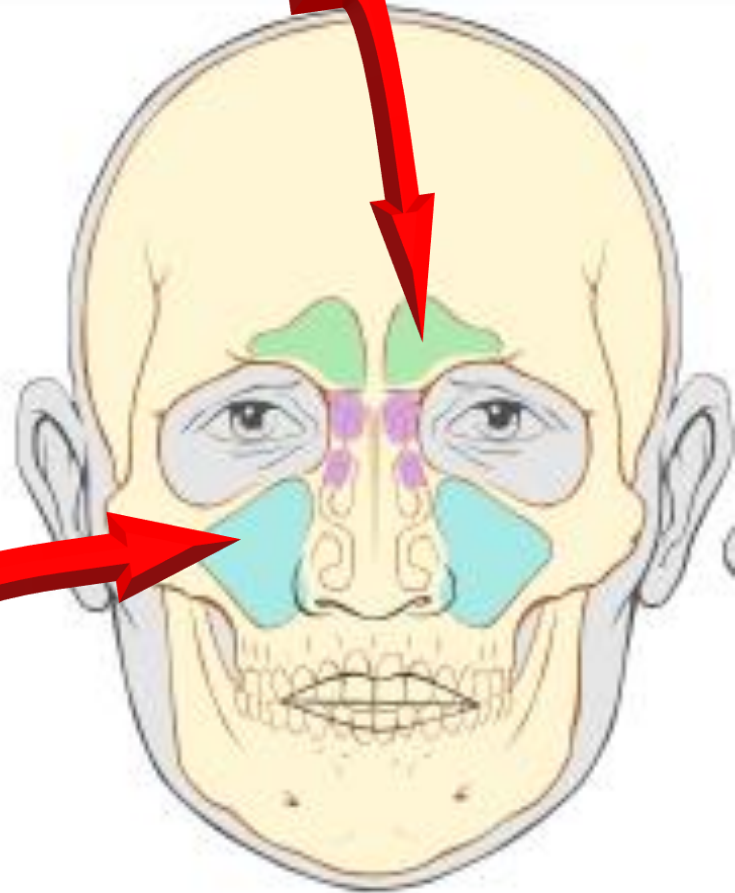
- 5) Pneumatic bones (air filled spaces inside some skull bones) →
maxilla &
frontal bones



A



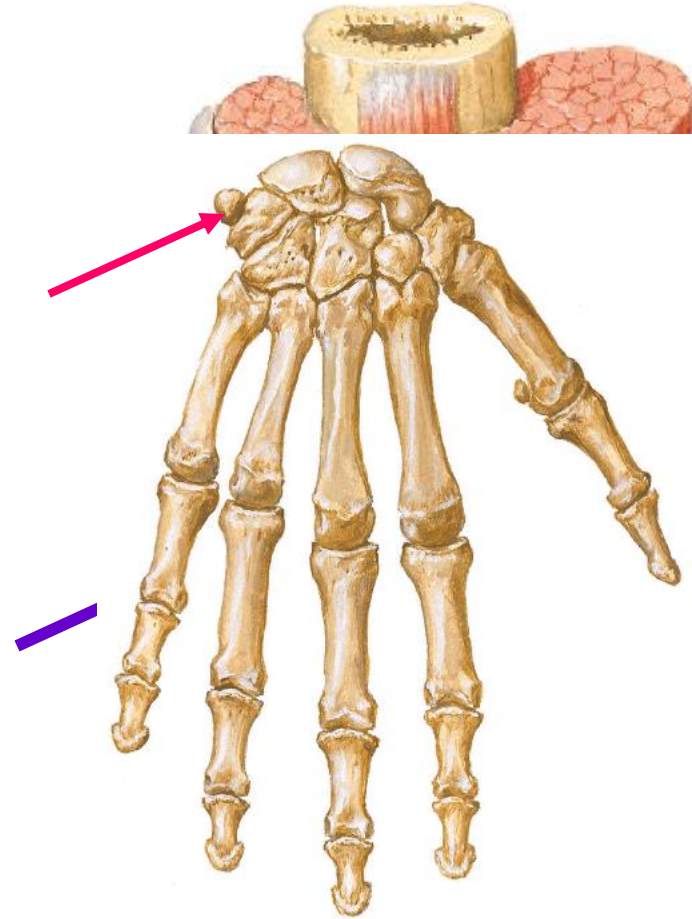
Pneumatic
bones
(air filled)

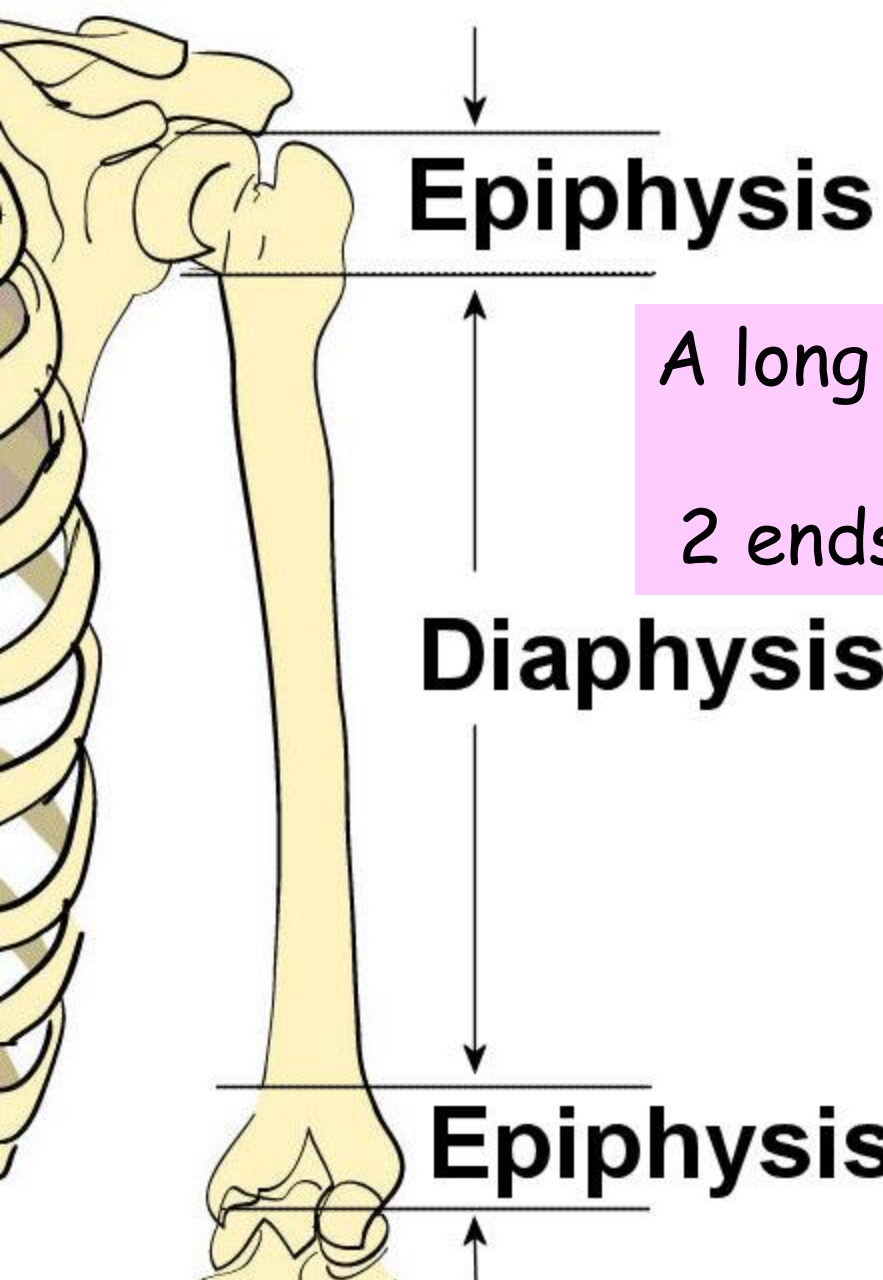


Pneumatic bones make **skull**
lighter & add resonance to
voice.



- 6) Sesamoid bone
(develop inside the
tendon of some
muscles at the
sites of friction) →
patella & pisiform
bone





Epiphysis

A long bone has a shaft (diaphysis)

2 ends (Upper & lower epiphysis)

Diaphysis

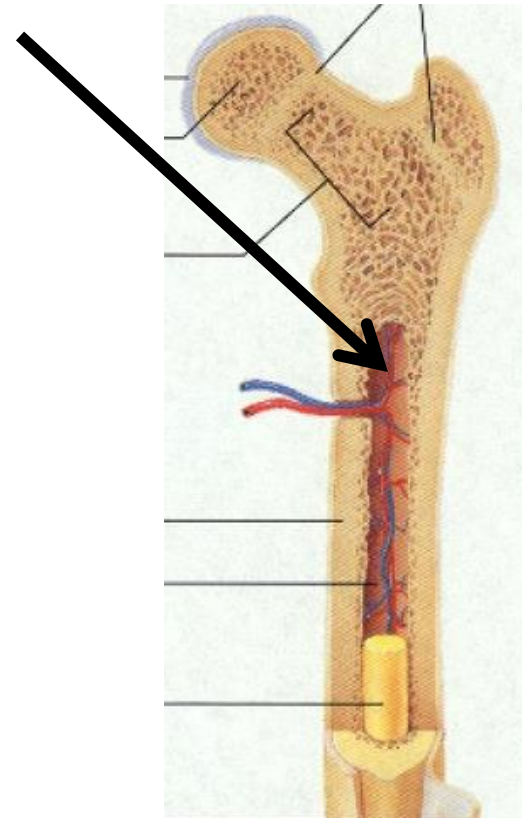
Epiphysis

Structure of a Long Bone

Structure of a Long Bone

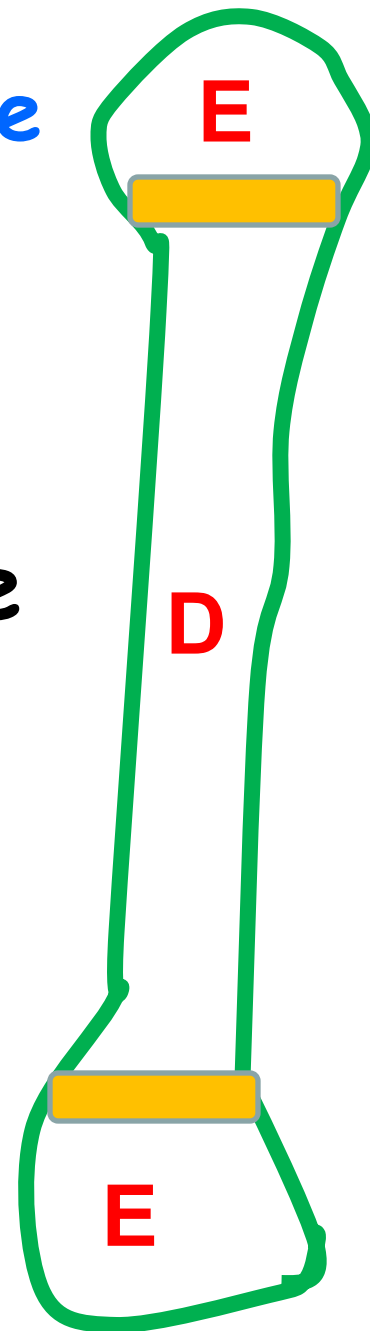
- **Diaphysis (Shaft)**

It contains a bone marrow cavity.



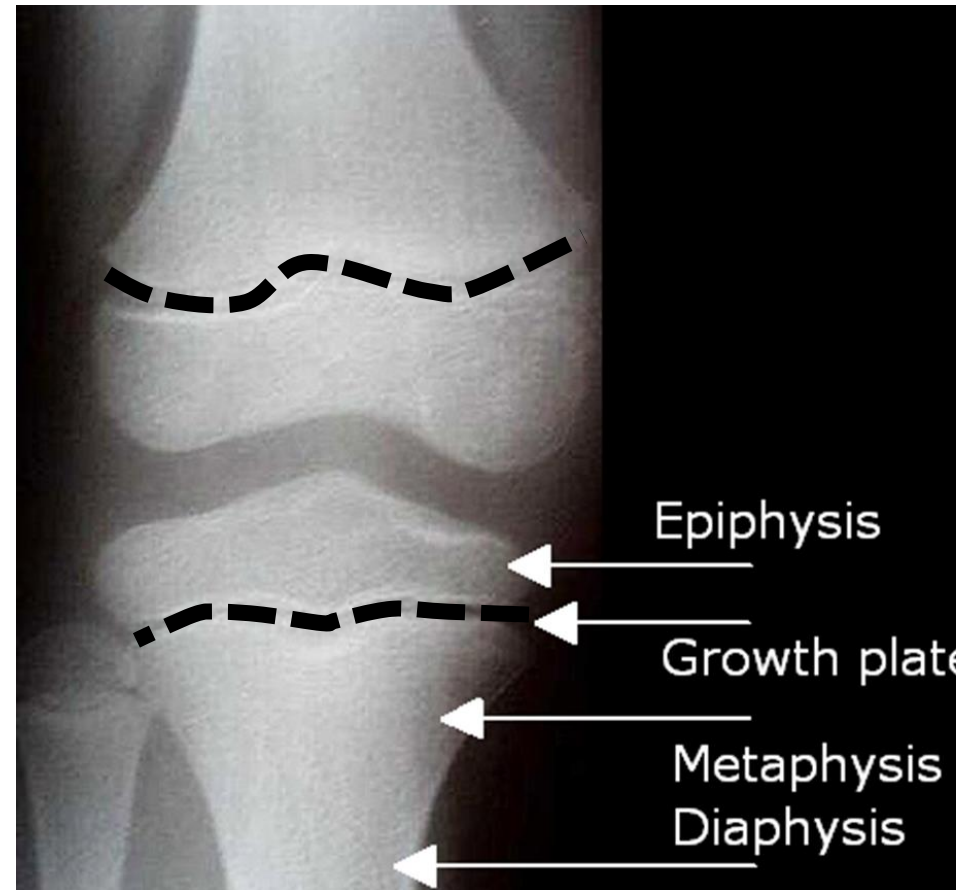
Epiphyseal plate of cartilage
Lies between
epiphysis & diaphysis
It proliferates to
increase **length** of bone

Epiphyseal plate
is responsible
for growth of
bone in **length**





**X-Ray of adult
knee bones**



X-Ray of young knee bones
The presence of an epiphyseal
Plate of cartilage indicates that
these bones will still grow in length

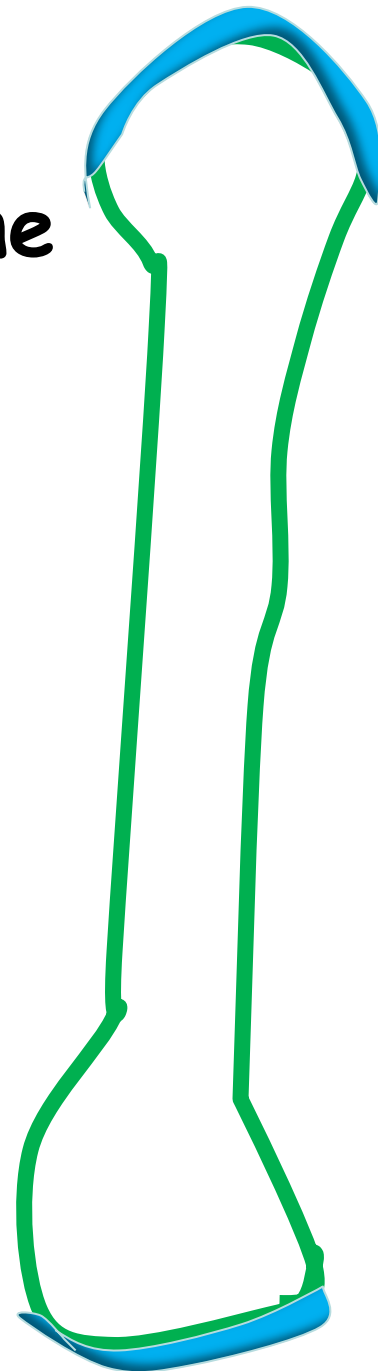
Articular Cartilage

Articular ends of a long bone are covered with articular cartilage

Why?

To provide a smooth surface where bones articulate together

to prevent friction between the articulating surfaces.

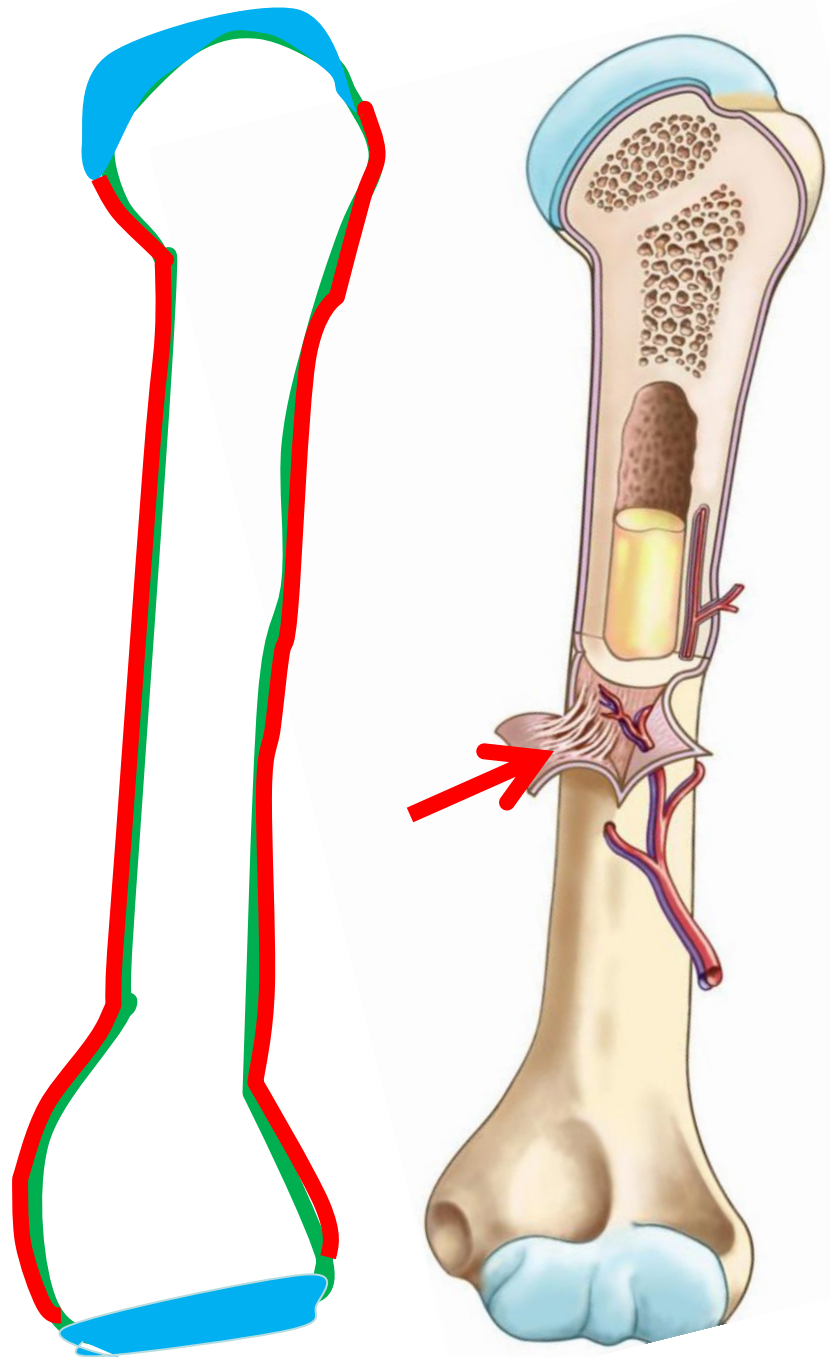


Periosteum

Peri = around
Osteum = bone

Membrane that covers surface of bone **except** its articular ends

Periosteum is rich in blood vessels

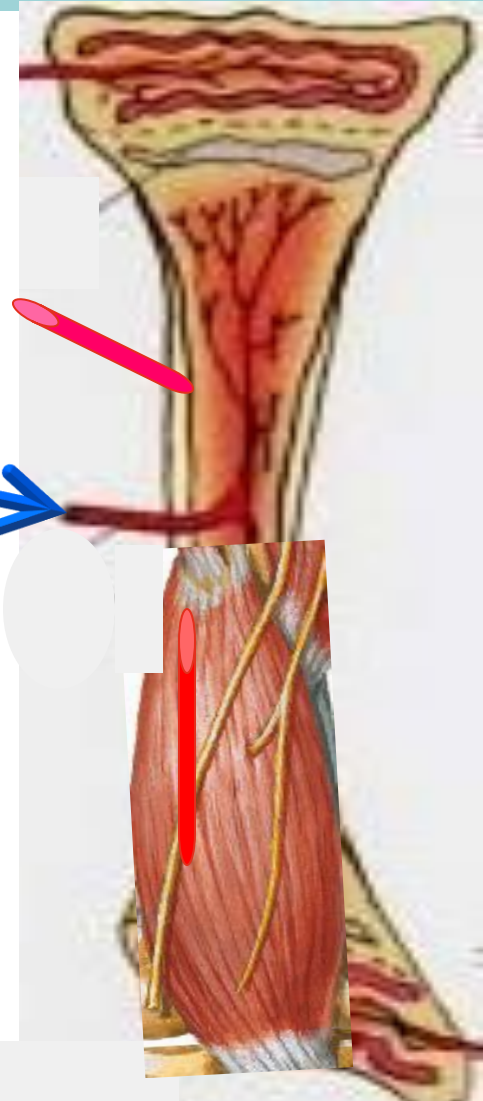


Functions of periosteum :

- Increases **breadth (diameter)** of bone by depositing new bone on surface
- Helps in repair of **fractures** of bone
- Supplies **blood** to the bone

Arterial supply of bones

- 1) Periosteal arteries
- 2) Muscular arteries
- 3) Nutrient artery:
enters shaft of
bone near its
middle through the
nutrient foramen .



Test yourself

- Which of the following bones belong to the appendicular skeleton or their girdles?
- A) skull
- B) sternum
- C) clavicle
- D) ribs
- E) vertebrae

• Only one of the following is an example of short bone:

- A) vertebrae
- B) phalanges
- C) patella
- D) Maxilla
- **E) carpal bones**

- In an X-ray ,the presence of an epiphyseal plate of cartilage (disc) indicates that:

- a. the bone length is increasing.
- b. the bone is fractured.
- c. the bone diameter is increasing.
- d. the bone is dead.
- e. the bone diameter is decreasing.

- Only one of the following bones develop by membranous ossification:
- A) vertebrae
- B) phalanges
- C) facial bones
- D) ends of clavicle
- E) carpal bones

Introduction to Human Anatomy

Anatomical terms L1

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ILOs

- **Describe the anatomical position and the anatomical planes**
- **Define the anatomical terms of position and movements**

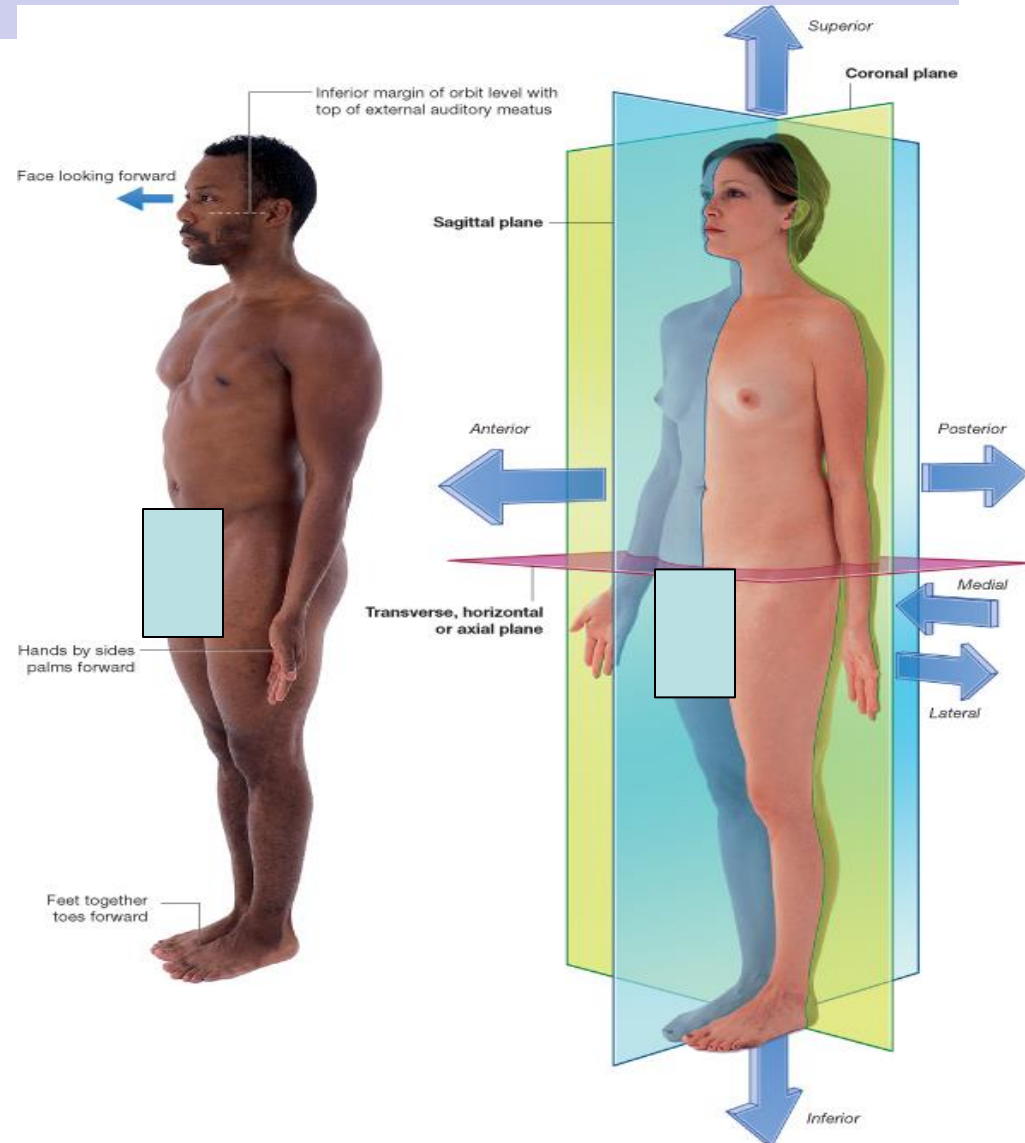
Anatomical Terminology

Anatomy:

Study of morphological structure of the body

Anatomical position

- Body is standing **erect**
- Face looking **forward**
- Upper limb **hanging** by side
- hands facing **forwards**
- Lower limbs are **parallel**



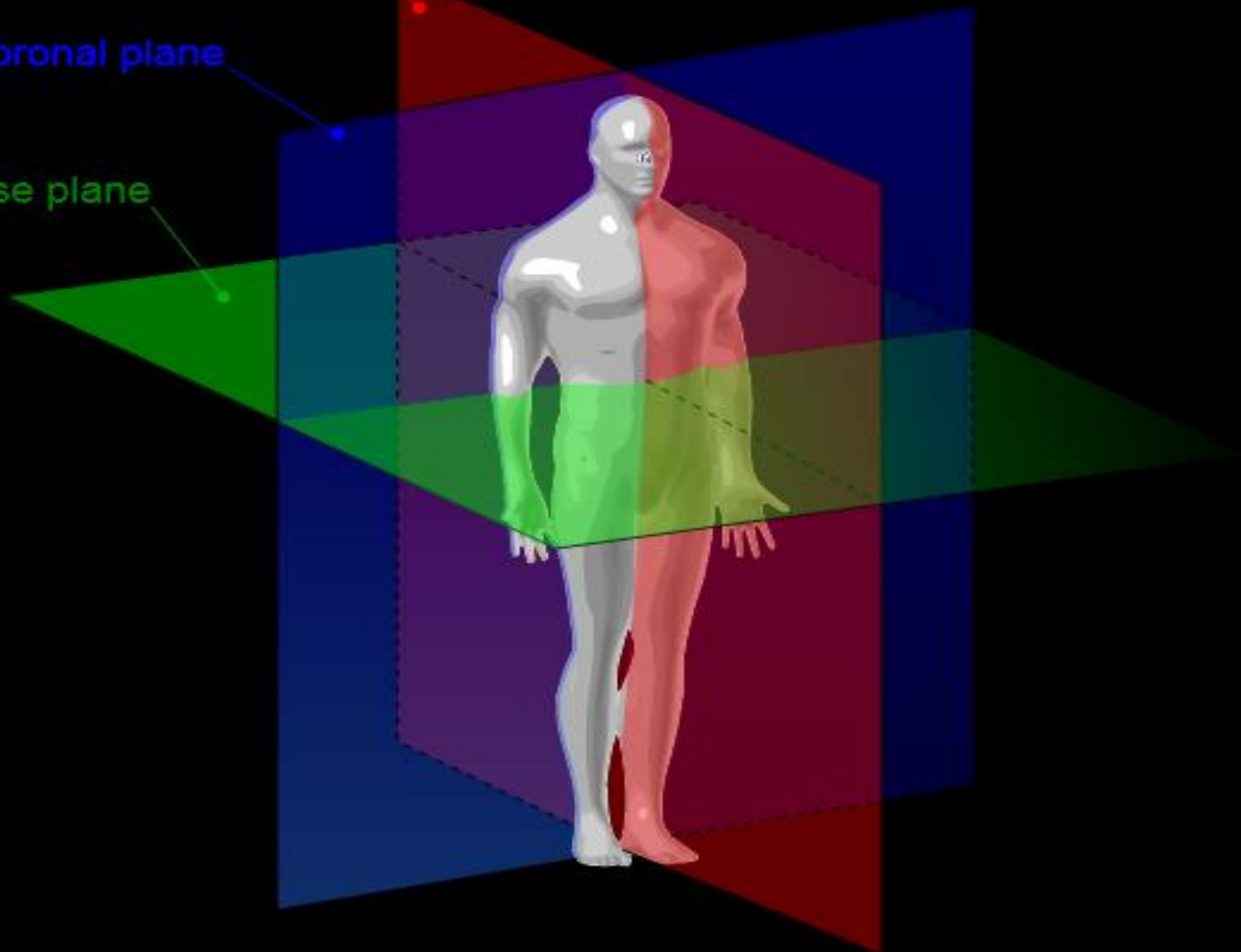
Anatomical Planes

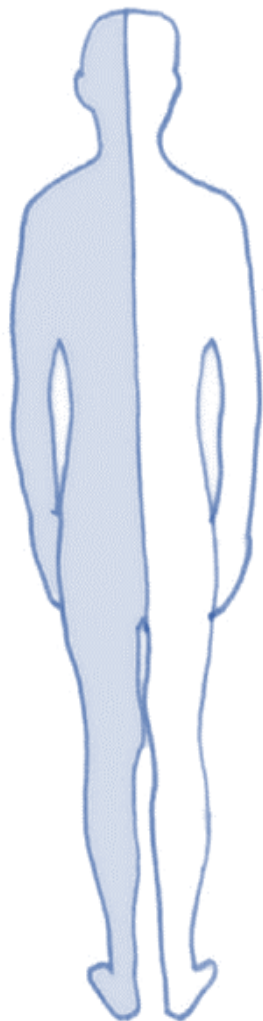
- **Median plane:** Vertical plane passing through center of body dividing it into **2 equal halves(right and left)**
- **Paramedian plane:** Vertical plane dividing body into **2 unequal halves**
- **Coronal plane:** Vertical plane perpendicular on median plane dividing body **into anterior& posterior**
- **Transverse:** Horizontal plane dividing body to **upper and lower parts**

Sagittal plane

Coronal plane

Transverse plane

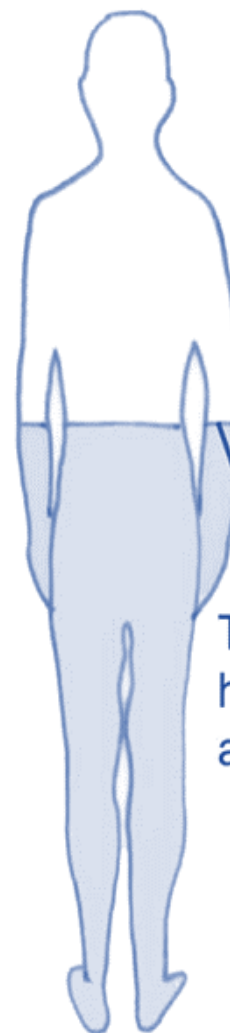




Median/Sagittal plane

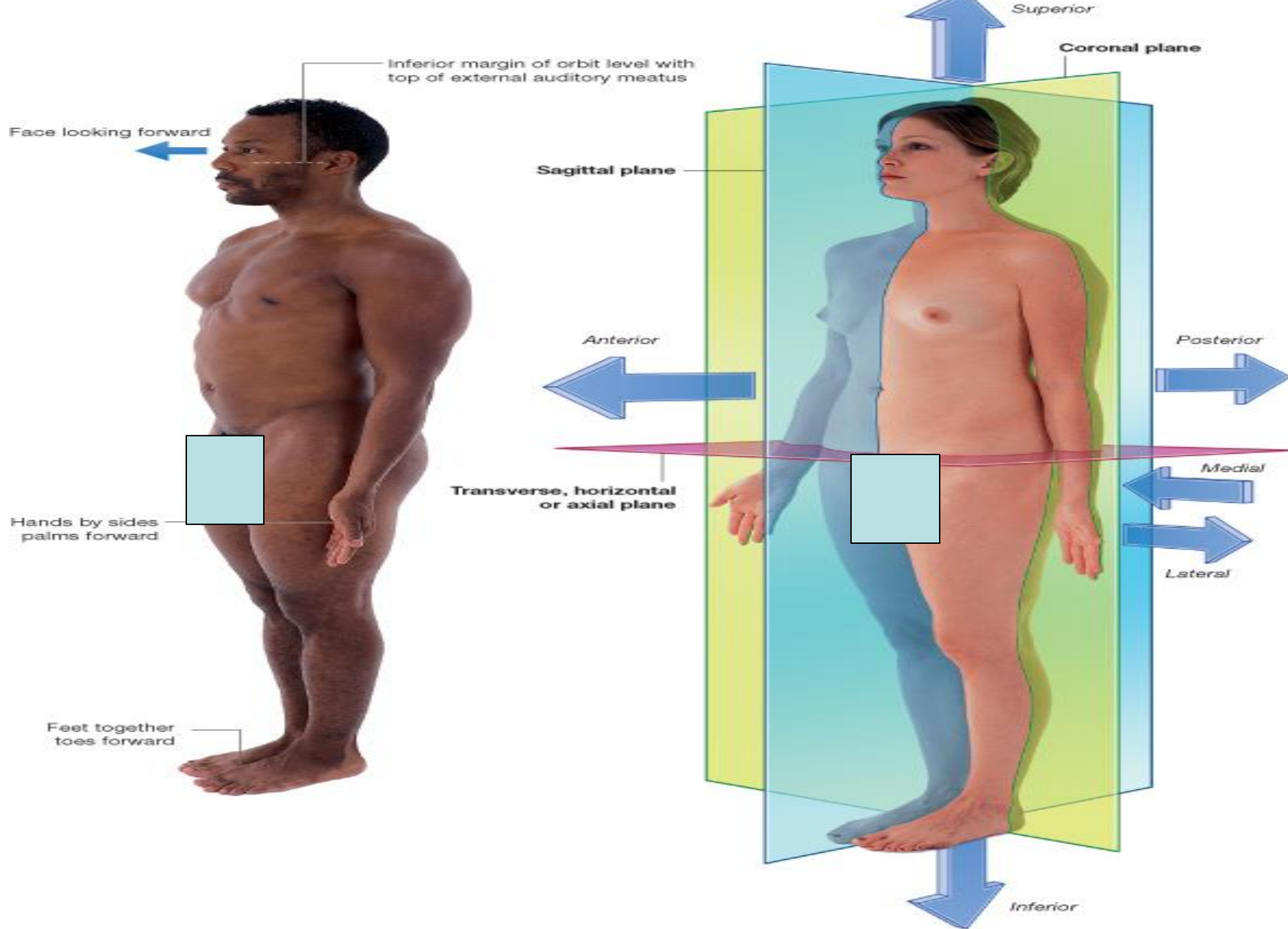


Frontal/coronal plane



Transverse/
horizontal
axis

Transverse/horizontal
plane



Anatomical Terms

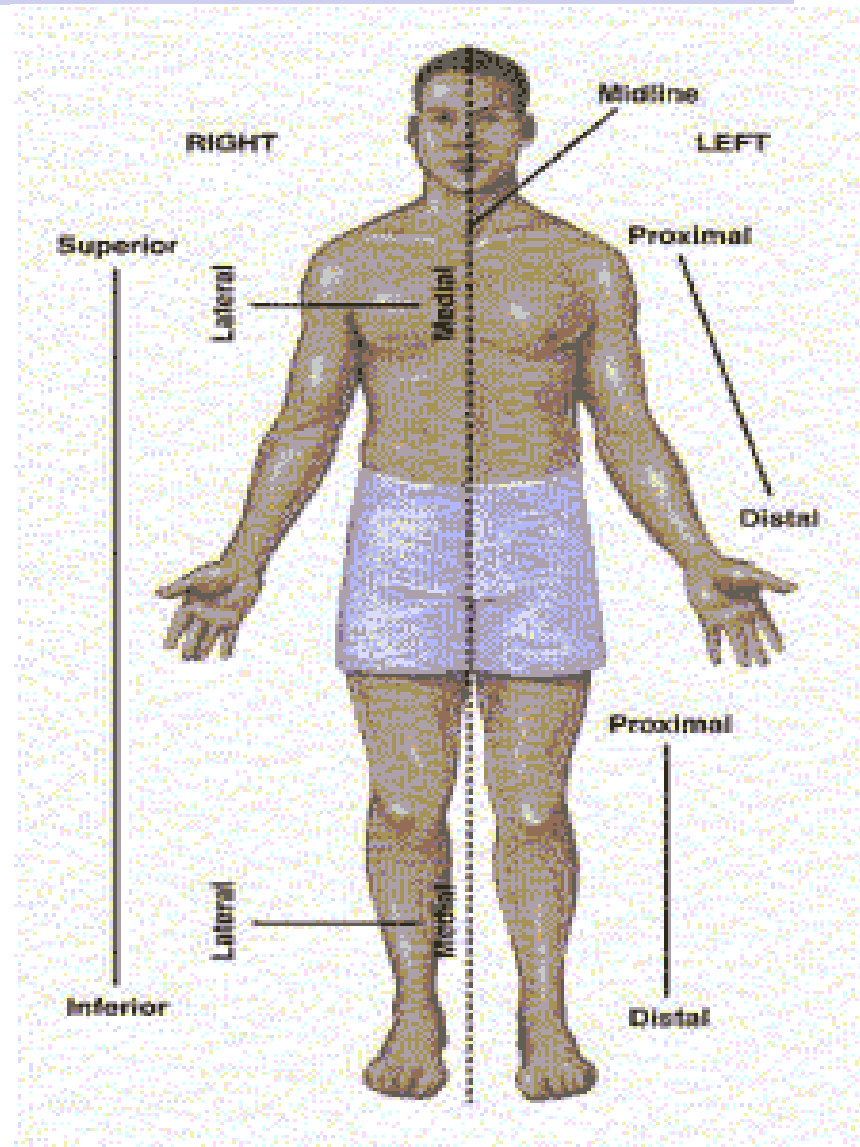
Median: structure lies **in** median plane

Medial : structure lies **nearer to** median plane

Lateral: structure lies **away from** median plane

Superior: close to upper end of body= skull

Inferior: close to lower end of body= floor



Anatomical Terms

- **Anterior:**

structure is nearer to **front**

- **Posterior:**

structure is nearer to **back**

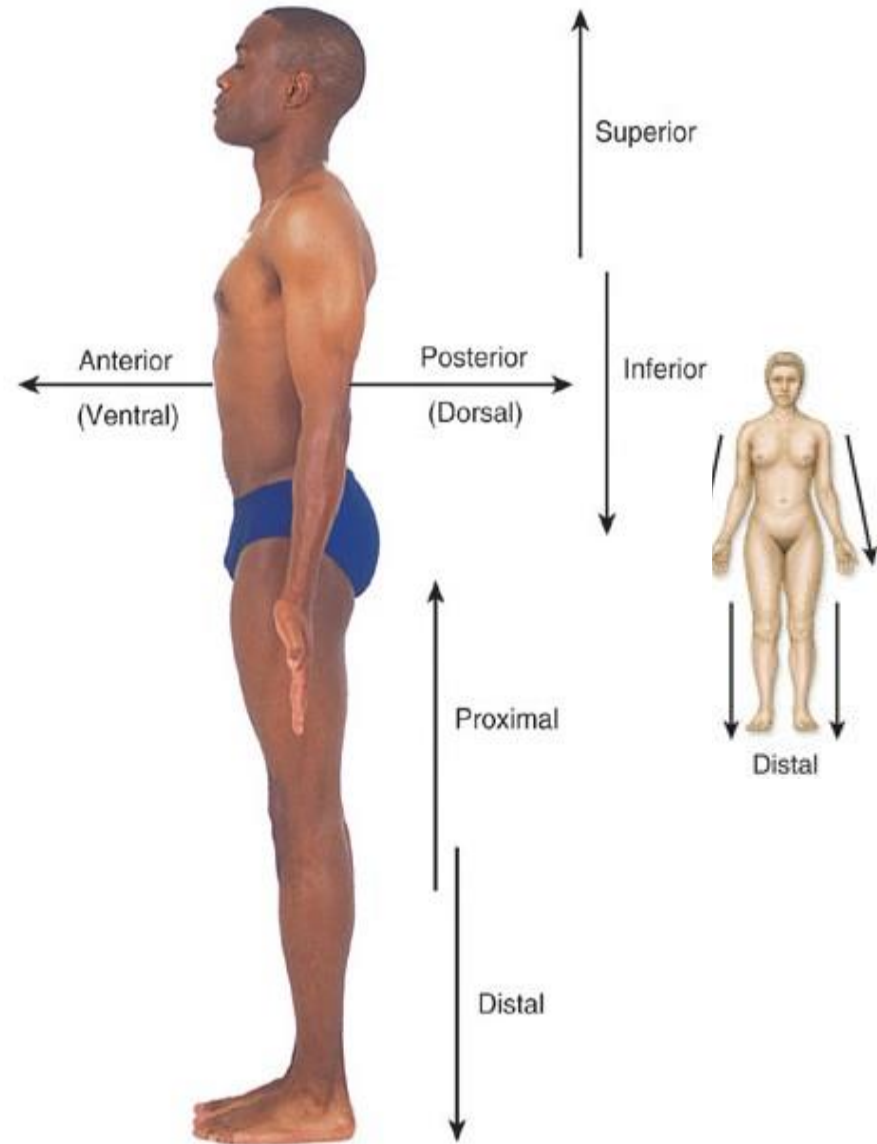
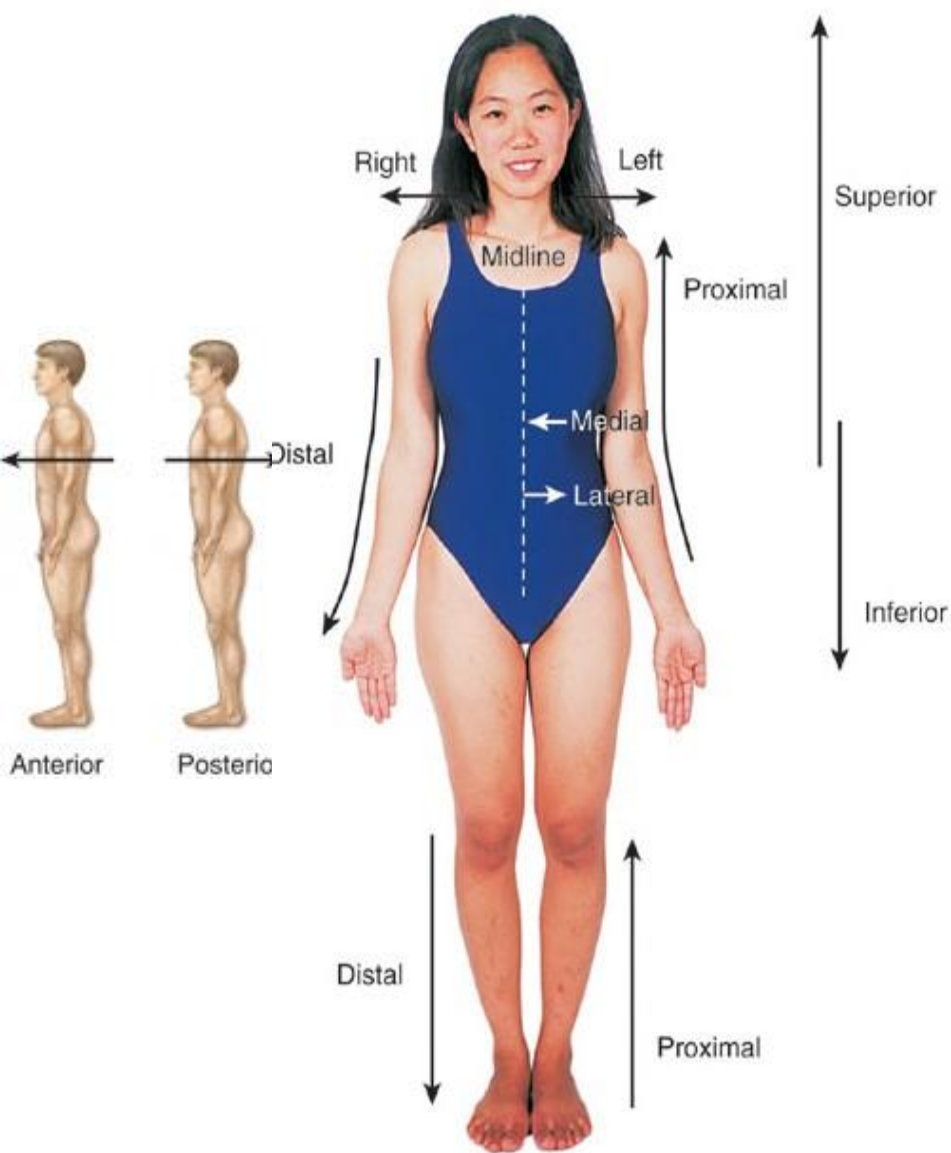
- **Superficial :**

structure **nearer to surface**

- **Deep:**

structure **away from surface**

- **Internal VS External :** structure is **inside** organ or cavity



Proximal:

Structure is nearer to trunk

Distal :

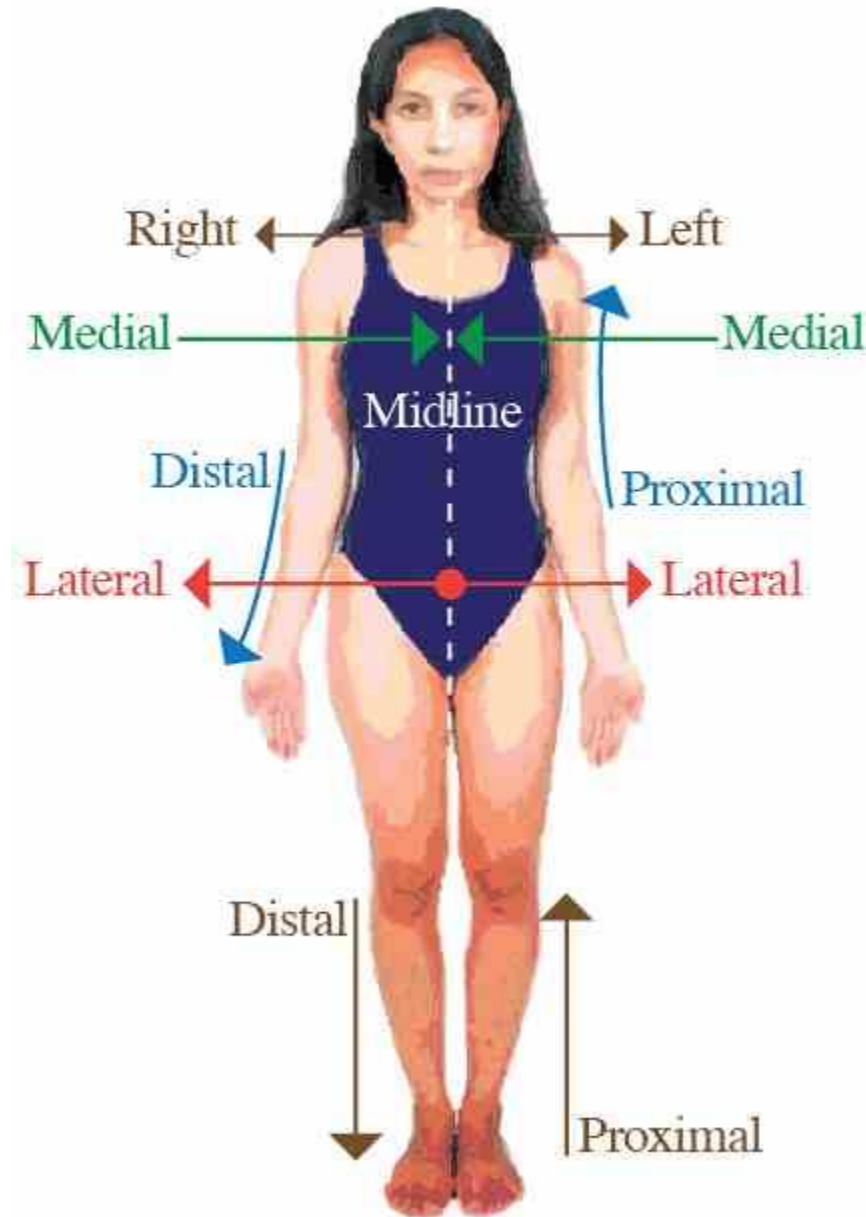
Structure is away from trunk

Ipsilateral:

Structure is on same side

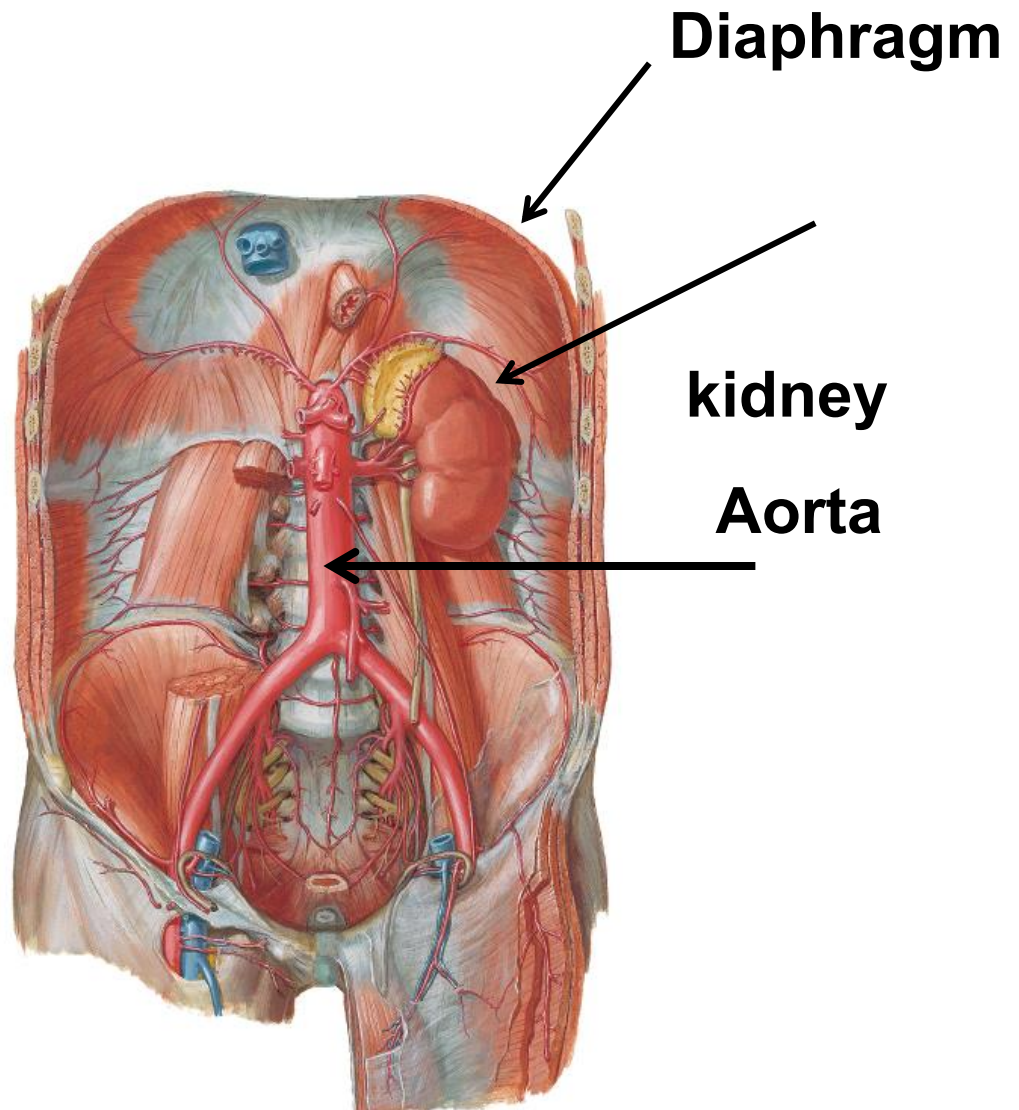
Contralateral:

Structure on opposite side

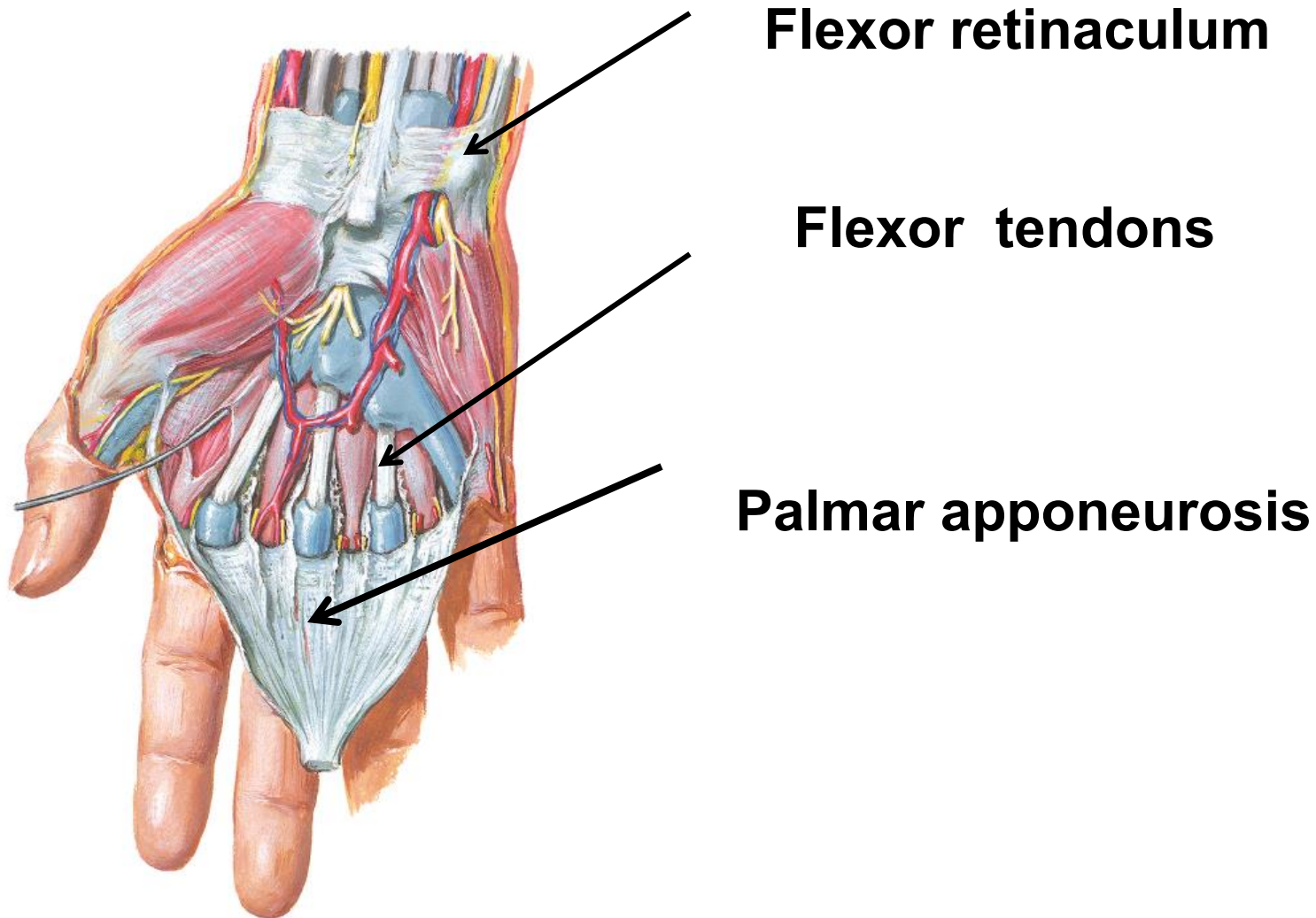


Interactive Exercise

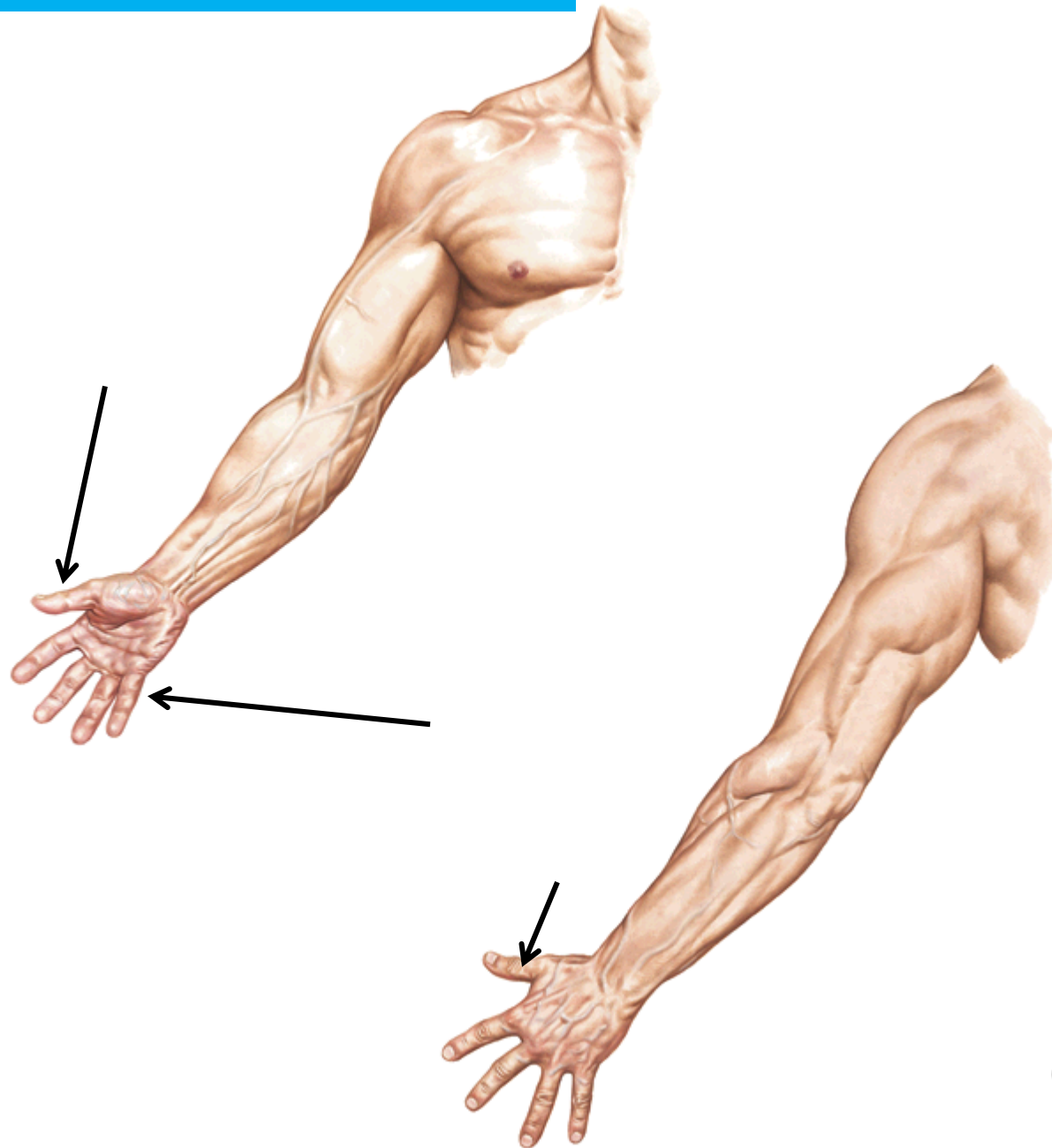
1. Which is medial kidney or aorta ?
2. What is superior to the kidney?



What is the relation of tendons to retinaculum

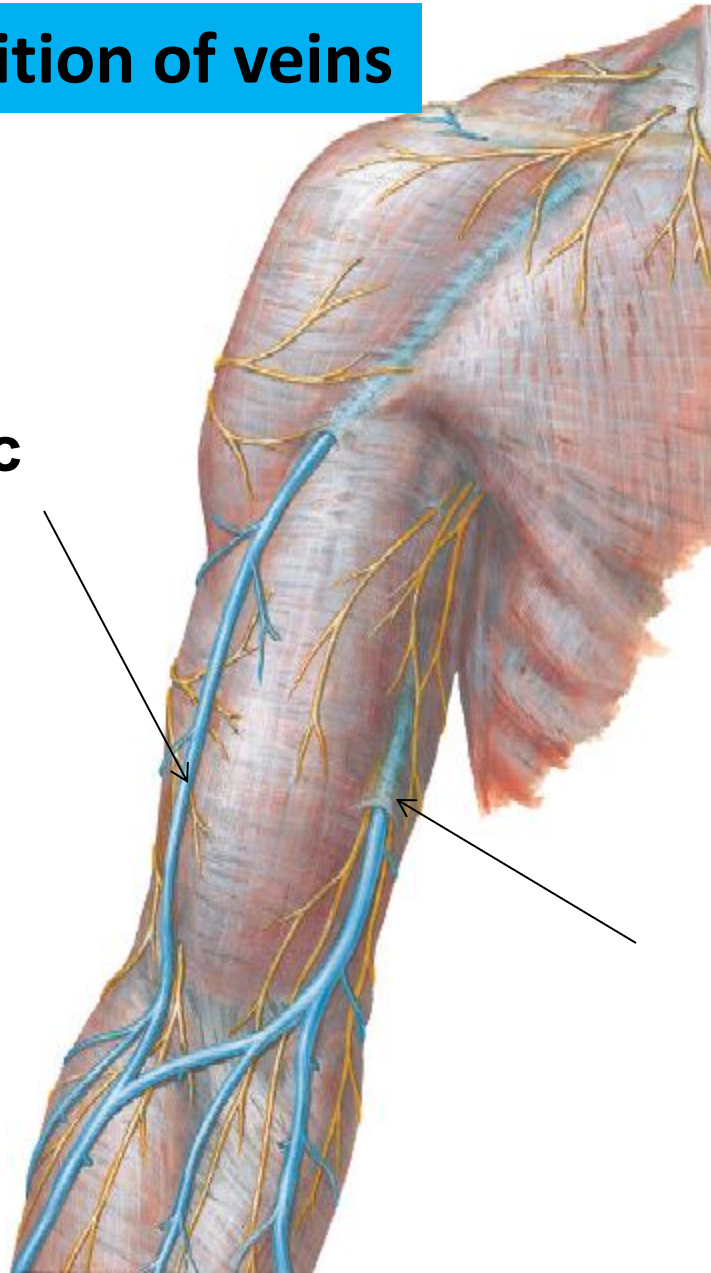


Which is medial thumb or little finger ?



Describe the position of veins

cephalic



basilic

Terms Used For Limbs

Terms used for limbs:

In the forearm:

radial or lateral & ulnar or medial.

In the hand:

palmar or anterior & **dorsal** or posterior.

In the leg:

fibular or lateral & **tibial** or medial.

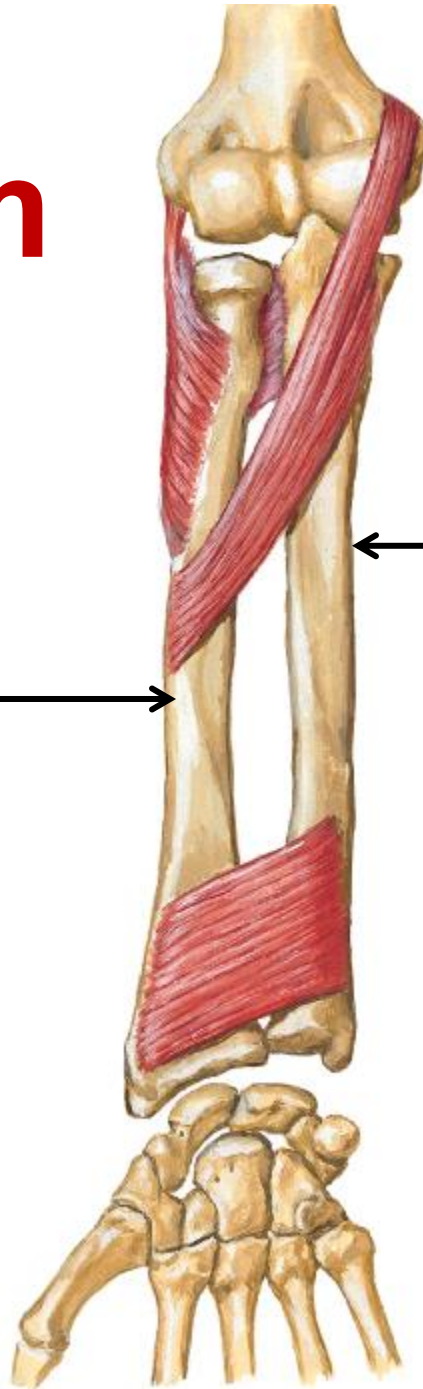
In the foot:

plantar or inferior & **dorsal** or superior.

Forearm

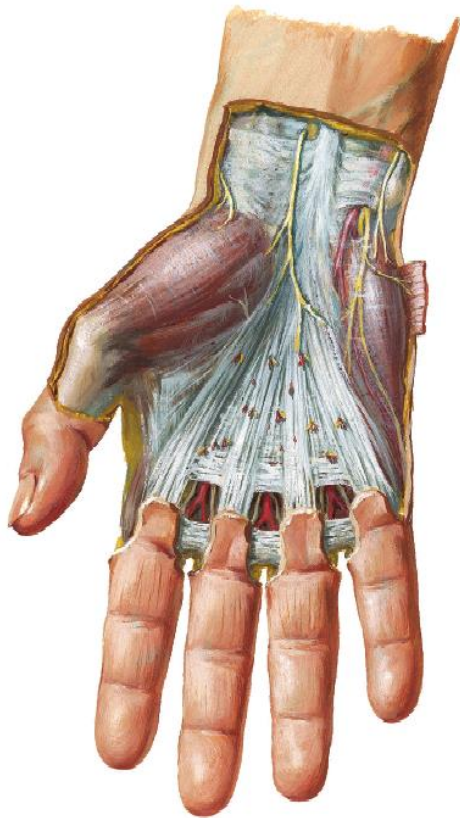
Radial or lateral

Ulnar or medial

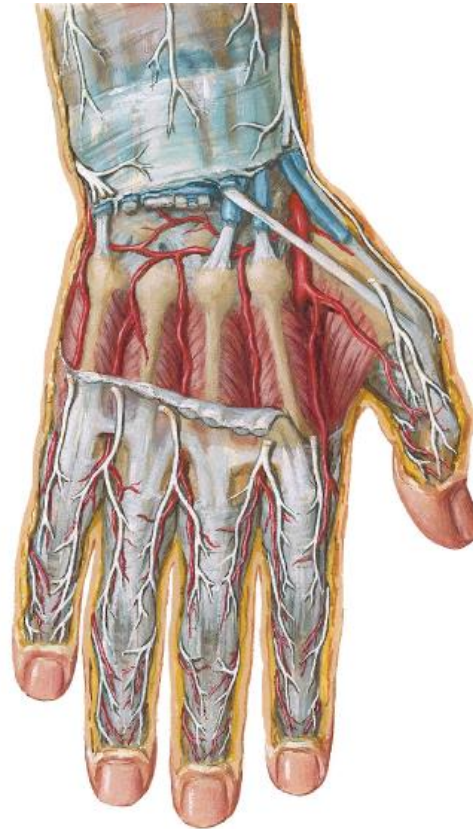


Hand

Palmar or anterior



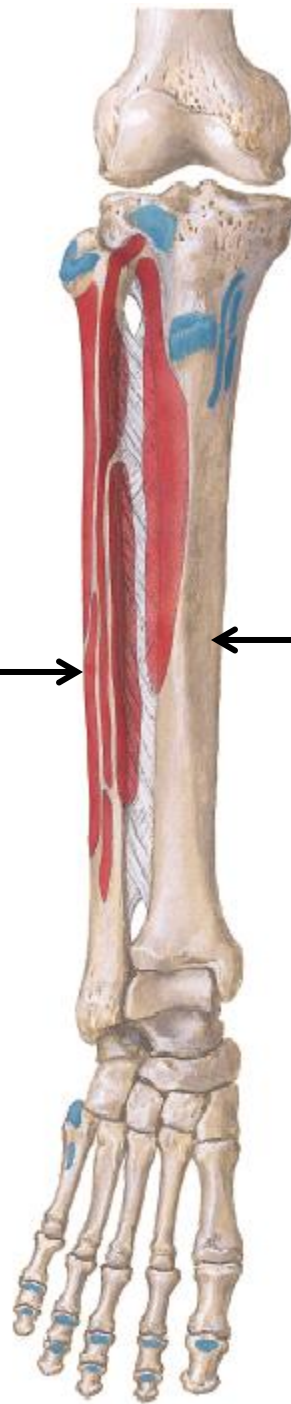
Dorsal or posterior



Leg

Fibular or lateral

Tibial or medial



Foot

Dorsum or superior



Planter or inferior



Terms of Movements

Flexion:

bending or approximation of two surfaces

Extension:

the **return back** of flexion

Abduction :

moving a limb **away** from the midline

Adduction:

moving a limb **towards** the midline

Adduction :of fingers

is approximating them **together**

Abduction: of fingers

is **spreading** them apart

TERMS OF MOTION



Flexion

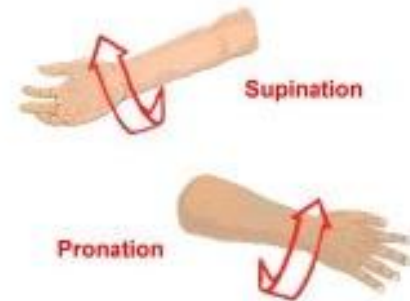


Extension



Medial Rotation

Lateral Rotation



Supination

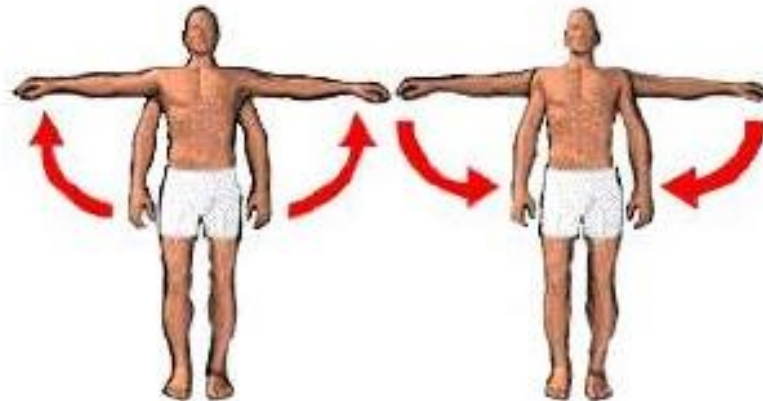
Pronation



Supine



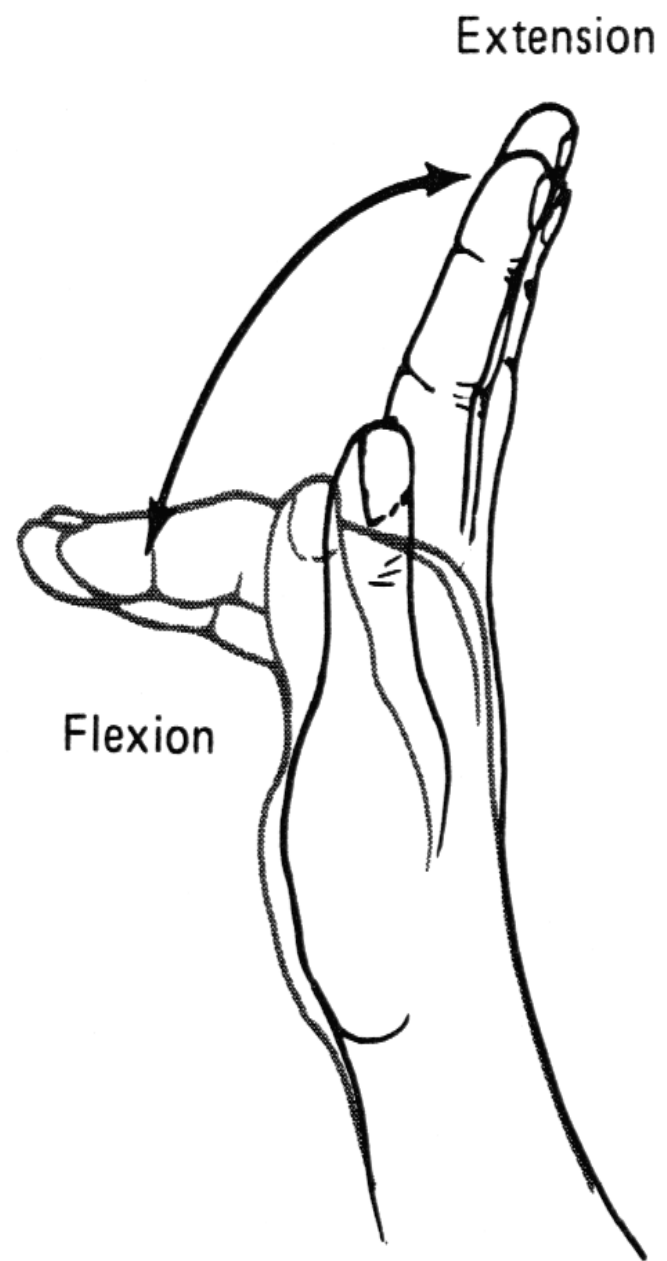
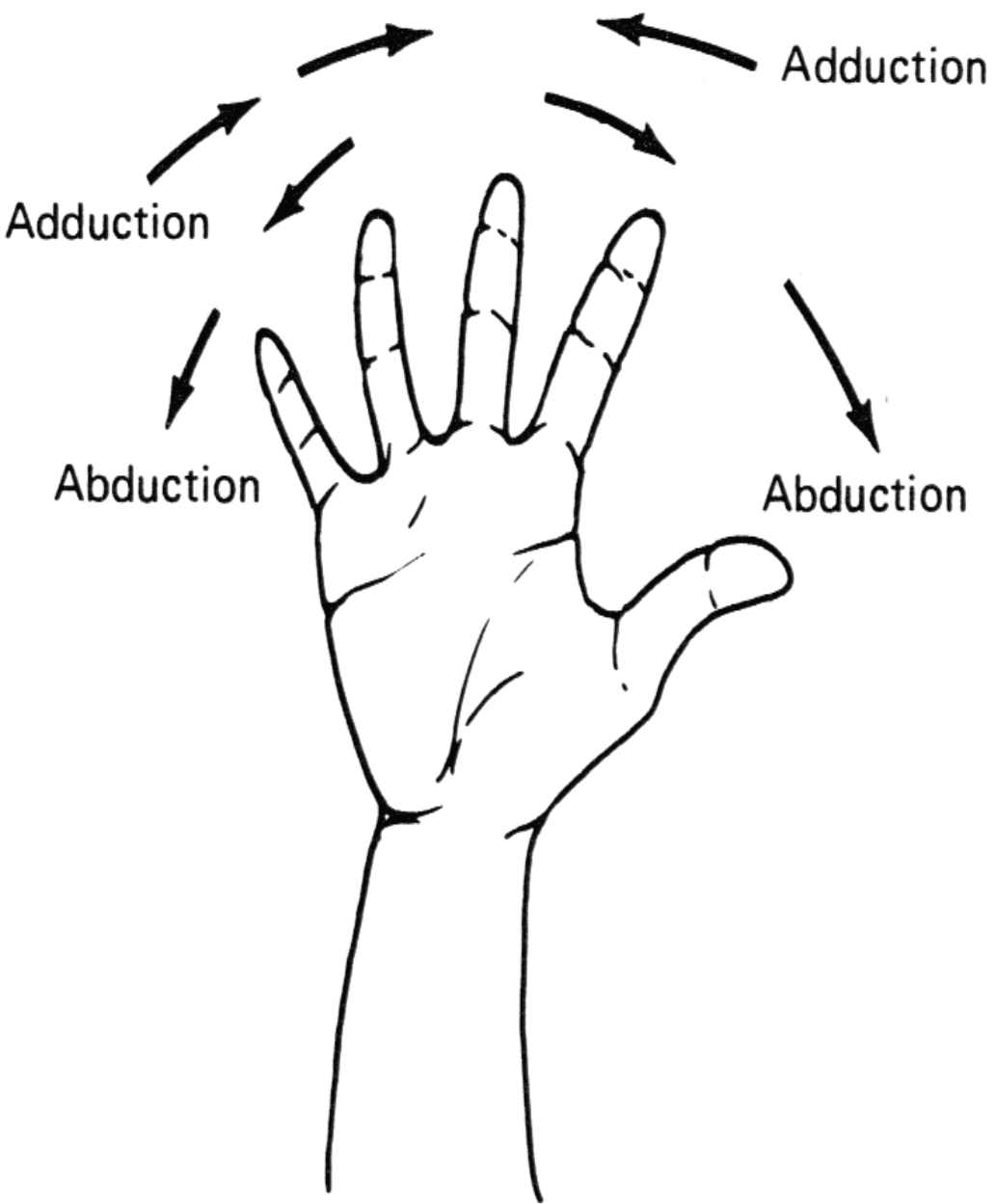
Prone



Abduction

Adduction





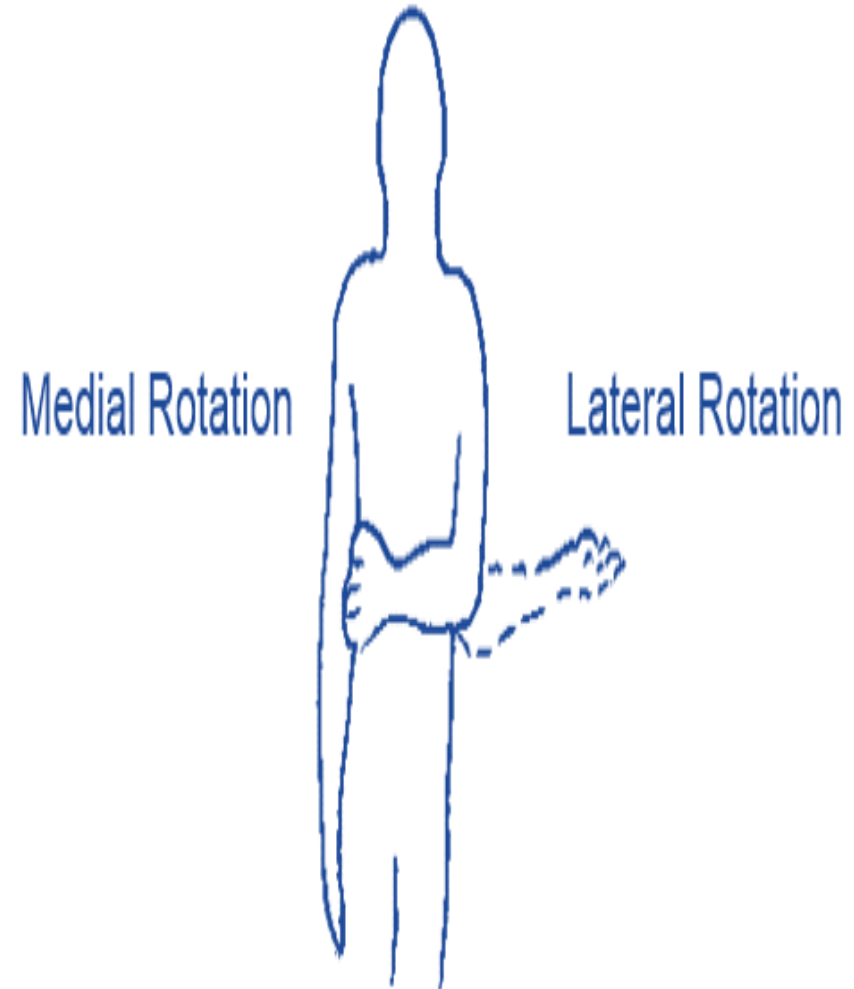
Terms of Movements

Medial rotation

turning the limb
along its
longitudinal axis
medially

Lateral rotation

turning the limb
along its
longitudinal axis



Rotation

- Medial rotation
- Lateral rotation

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(a) Medial (internal) rotation



(b) Lateral (external) rotation

Pronation

brings the palm of hand backward

Supination

brings the palm of hand forwards

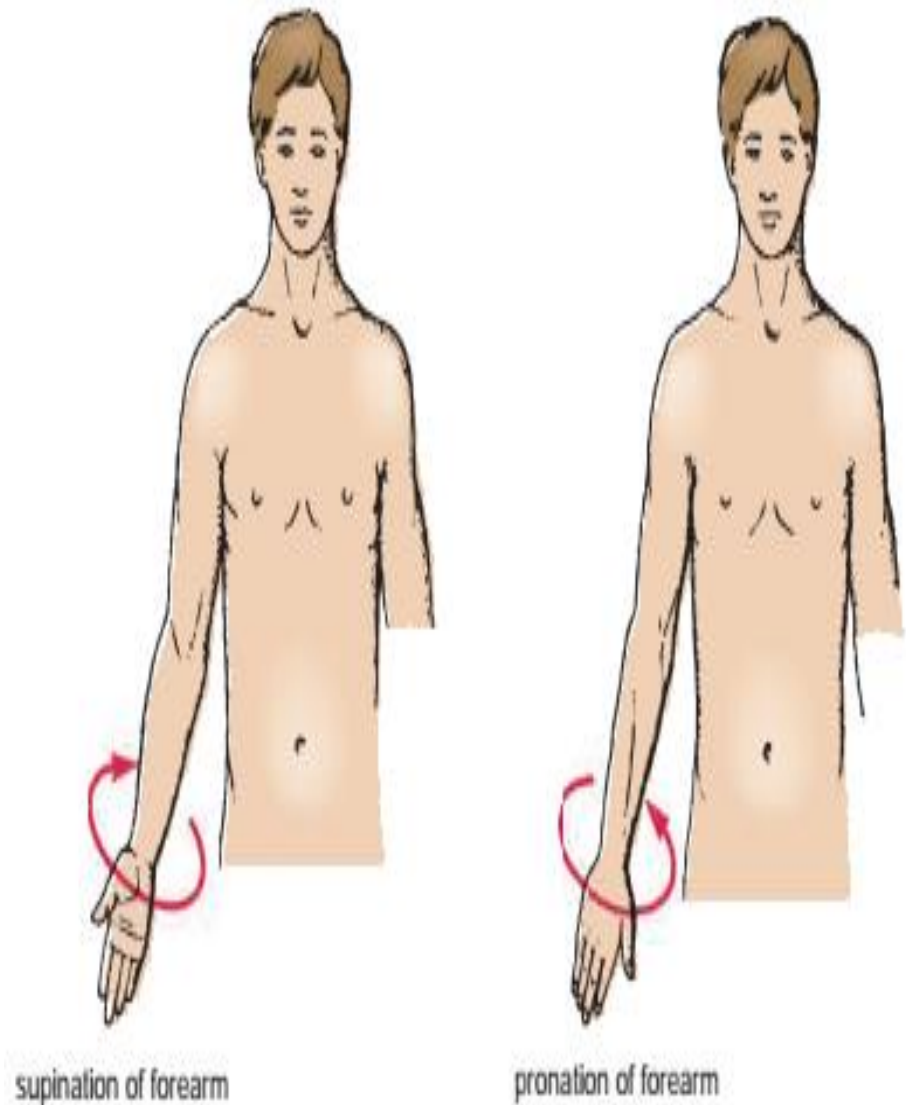


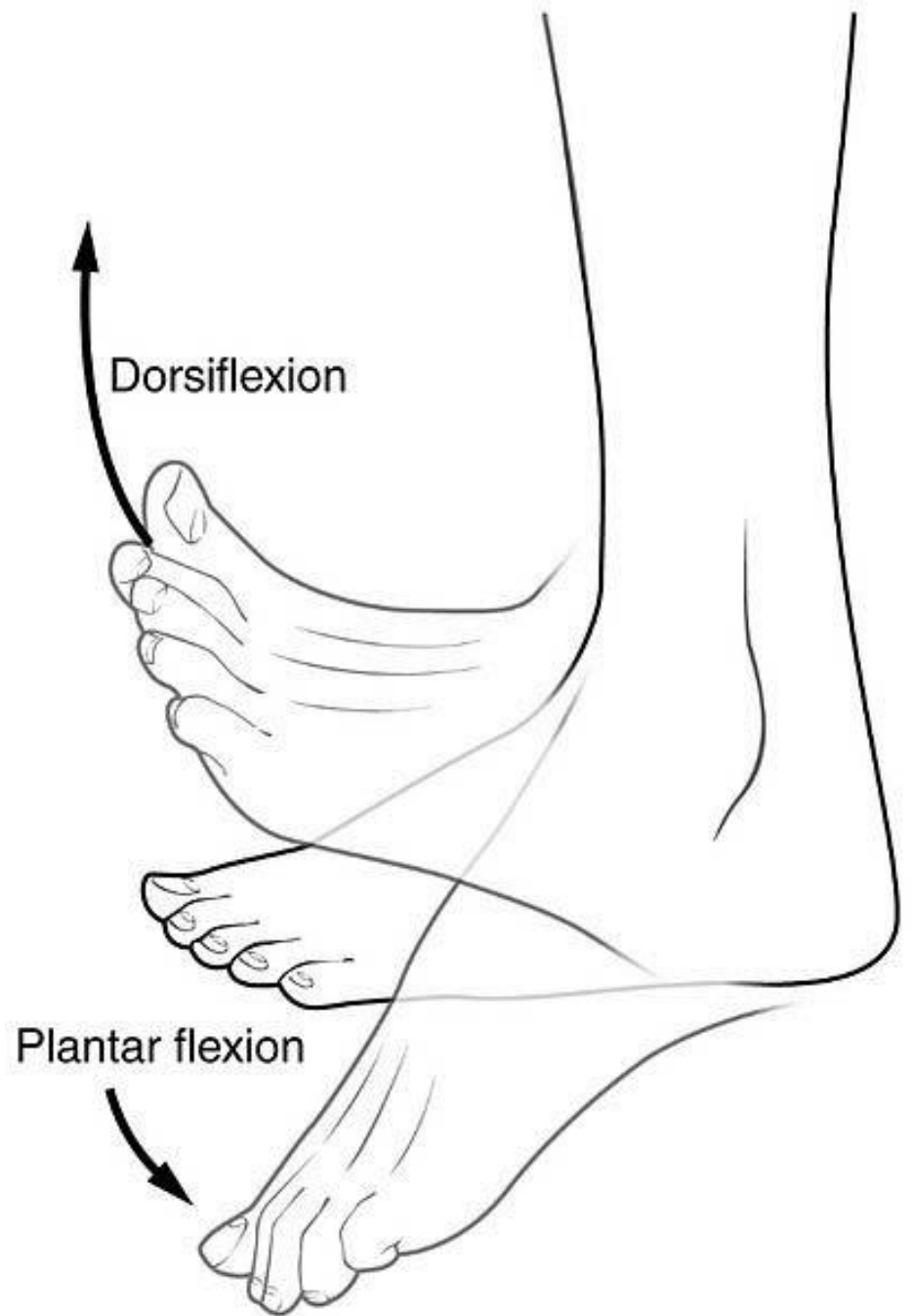
Figure 16

Dorsiflexion

bending the foot
at the ankle
upwards

Plantar flexion

bending the foot
at the ankle
downwards



Inversion

rotating the foot so
that the sole faces
inwards

Eversion

rotating the foot so
that the sole faces
outwards



Special Movements

- Opposition
 - Touching the thumb to each finger.





flexion of leg



extension of leg



adduction of thigh



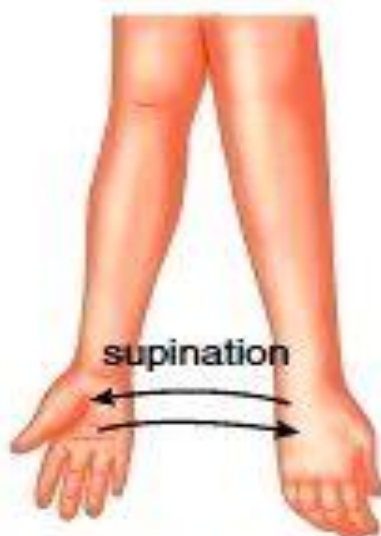
abduction of thigh

a. Angular movements



rotation
of arm

b. Circular movements



pronation of hand



inversion



eversion

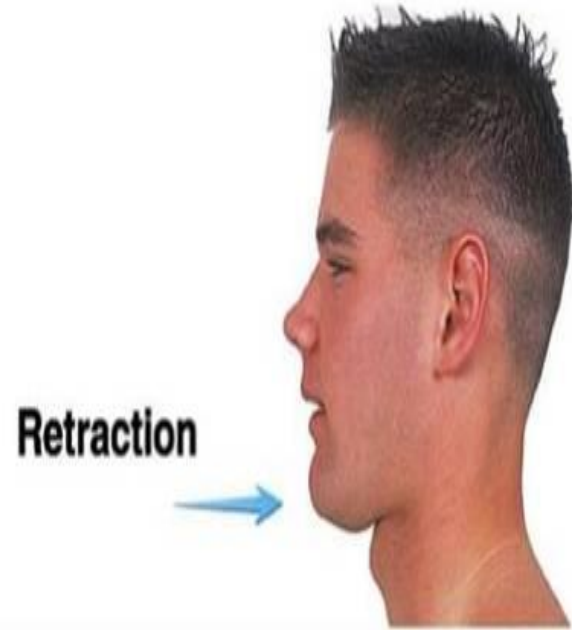
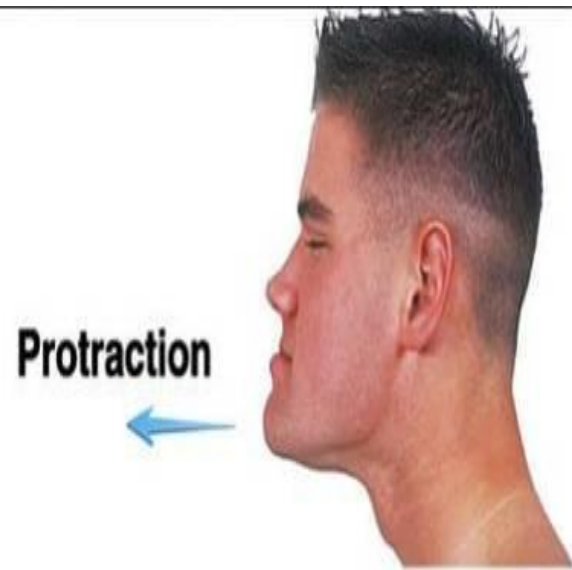
c. Special movements

Protraction

moving a
part forward
(protraction of
mandible or shoulder)

Retraction

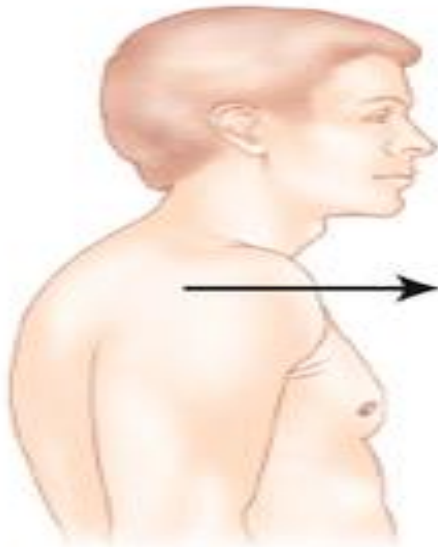
moving a part
backward
(retraction of
mandible or
shoulder)



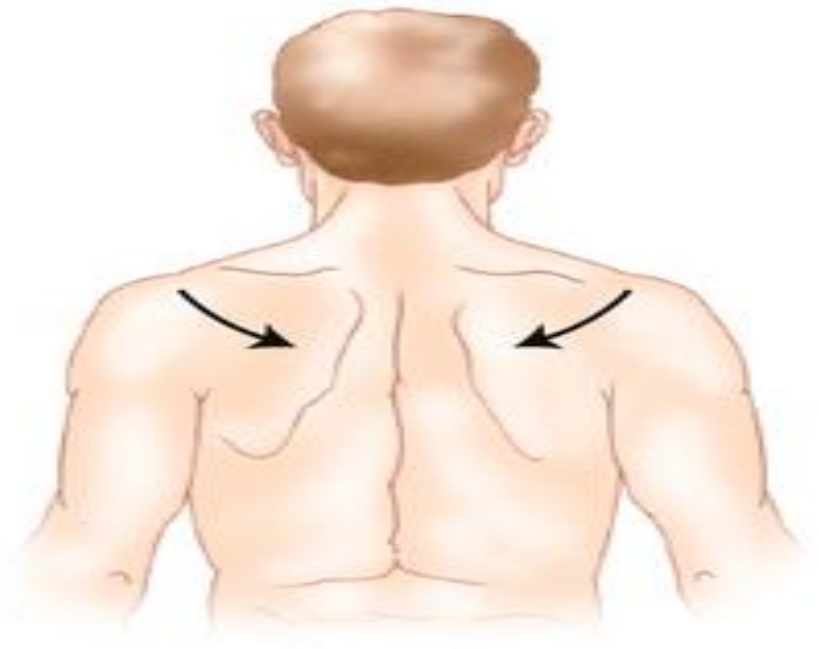
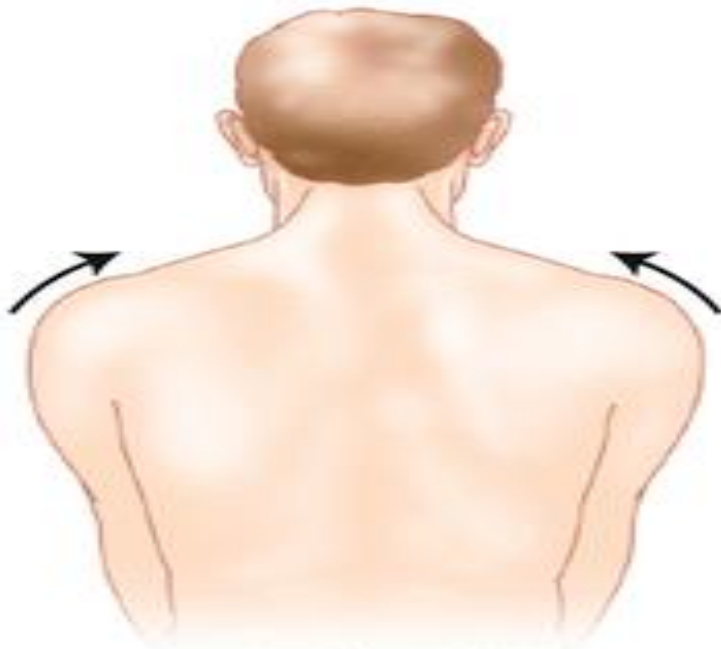
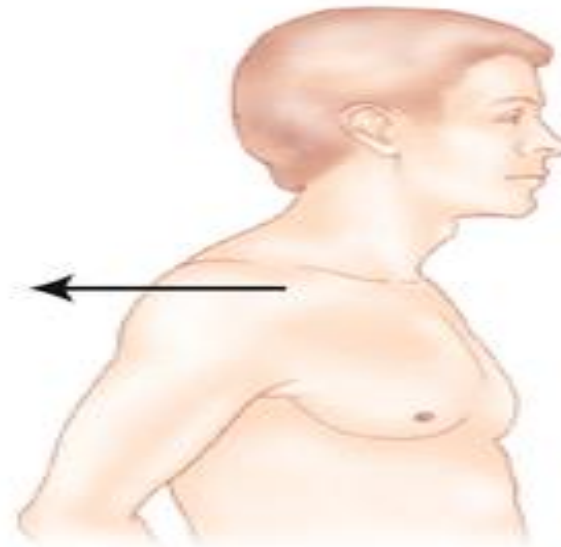
Shoulder Protraction



Protraction



Retraction



Elevation

moving a part upward

(elevation of shoulder or mandible)

Depression

moving a part downward

(depression of shoulder or mandible)

Elevation



Depression





Thank
you!!